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REVIEW

Reported seaweed diseases in aquaculture: a systematic review

Ade Wahyul Fajri Alimin[1,9] · Kasmiati Kasmiati[2,9] · Aswin Aswin[3] · Fatchurrachman Fatchurrachman[4] · Nur Yanti Pangloli[5] · Dea Pramita[2] · Winda Wijaya[6] · Rahman Romli[7] · Elmi Nurhaidah Zainuddin[2,9,10] · Jamaluddin Fitrah Alam[8,9]

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Abstract

Since the late 1980s to 2025, at least 16 disease entities have been reported to affect commercially cultivated seaweeds, occurring in both open-water farming systems and nursery facilities. The expansion and intensification of seaweed production have increased the frequency of host–pathogen interactions, a trend further exacerbated by environmental stressors associated with climate change, including ocean warming, salinity fluctuations, and acidification. These conditions have contributed to rising disease incidence, with consequences such as reduced yields and product quality, greater economic losses, and a potential weakening of the overall resilience of the seaweed industry. This review was conducted using a systematic literature review approach by compiling and screening scientific publications from Scopus, Web of Science, SpringerLink, and two additional relevant sources to provide a comprehensive synthesis. Based on the evidence assembled, this review aims to (i) map the relationships between biotic and abiotic factors and the emergence and spatial distribution of diseases across seaweed species and cultivation regions, (ii) examine the economic consequences of disease along the seaweed value chain, and (iii) evaluate key aspects of disease management in seaweed aquaculture systems. Overall, this review provides a scientific basis for shaping research and policy agendas that prioritise risk-based biosecurity, the development of tolerant varieties, early-warning systems and rapid diagnostics, high-resolution environmental monitoring, and strengthened collaborative governance through public–private partnerships to support sustainable and disease-resilient seaweed aquaculture.

Keywords Seaweed aquaculture · Diseases · Environment · Pathogens · Economic burden · Risk-based biosecurity

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Extended author information available on the last page of the article

Introduction

The global seaweed cultivation sector has expanded in response to growing demand from various industries, such as food, cosmetics, pharmaceuticals, biomedicine, dermatology, agriculture, and fisheries (Polat et al. 2023; Debbarma et al. 2024). Currently, the largest reference database for seaweed taxonomy (www.algaebase.org) recognises over 12 thousand seaweed species (Vigil et al. 2024), with more than 291 species being commercially cultivated worldwide (Iqbal 2022). Among these, brown seaweeds (*Laminaria/Saccharina*, *Undaria*, *Sargassum*), red seaweeds (*Kappaphycus* sp., *Eucheuma* sp., *Gracilaria*, *Pyropia*/*Porphyra*), and green seaweeds (*Ulva* and *Caulerpa*) are the most commonly farmed species (Iqbal 2022). In 2021, global seaweed production amounted to 35.1 million tonnes in live weight, with 94% of the total yield concentrated in four primary countries: China contributed 60%, Indonesia 25%, South Korea 5%, and the Philippines 4% (UNCTAD 2024). Embracing this development, the global seaweed market is projected to reach USD 17.8 billion by 2032 (Spagnuolo and Genovese 2024). Furthermore, seaweed is projected to play a crucial role in meeting the needs of a global population expected to exceed 9 billion by 2050 (Duarte et al. 2022). However, based on the systematic literature review method we applied, it is evident that numerous reports of disease incidents have been published from the late 1980s to 2025 (Table 1), reflecting a wide range of reported symptoms (Table 2). Diseases caused by pathogens such as bacteria, parasites, and viruses are becoming more prevalent, prompted by environmental stressors like fluctuations in salinity and rising seawater temperatures (Kim et al. 2023; Dekker et al. 2024; Zhuang et al. 2024). These stressors compromise the seaweed's capacity to produce antimicrobial agents, leaving it more susceptible to colonisation and infection by opportunistic pathogens (Largo et al. 2024). Furthermore, as cultivation intensifies and expands, the incidence and diversity of diseases in farmed seaweed species have increased and are widely documented in various regions (Fig. 1).

Despite growing concerns over disease outbreaks, several countries continue to set ambitious targets to become global leaders in seaweed production, recognising its potential as a key strategy for sustaining the sector and contributing to global food security over the next 30 years (Bhuyan 2023). Culinary demand spans taxa across climates, including *P. palmata* (traditionally eaten as dulse in the UK), *S. japonica* (haidai in Chinese or kombu in Japanese), *U. pinnatifida* (wakame), *Alaria esculenta* (alaria), *Pyropia* spp. (nori), *Sargassum fusiforme* (hijiki), and *C. lentillifera* (umibudō) (Mukhia et al. 2024). In tropical waters, beyond *C. lentillifera*, one of the most commercially prominent commodities is *Kappaphycus* spp., a carrageenophyte eucheumatoid, is widely employed as a thickener and gelling agent in jelly confectionery, candied products, crackers, jams, and seaweed meatball-like product (Poeloengasih et al. 2014). In line with these opportunities, Indonesia aims to dominate the global edible seaweed market through the roadmap set by Presidential Decree No. 33/2019, with eucheumatoids as the priority (Jompa et al. 2024). Concurrently, Malaysia revitalises aquaculture by promoting seaweeds as a primary future source of food and alternative protein (Ismail et al. 2024). This potential is further validated by evidence highlighting the high essential amino acid content in red seaweed (*Gracilaria*), the high polyunsaturated fatty acid levels and low saturated fatty acid content in brown seaweed (*Sargassum*) (Chin et al. 2023), and the significant protein levels in red seaweed (*Pyropia*), a widely cultivated species in temperate zones, which are comparable to the standards set by the World Health Organization (WHO) and the Food and Agriculture Organization (FAO) (Machado et al. 2020).

Table 1 Diseases and causative agents in aquaculture seaweed species worldwide

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
Caulerpa racemosa	Coastal area of Bangladesh	IID	n.o	Ma, Mi (LM)	Cultivated in open waters	Sarker et al. (2021)
Enteromorpha	Coastal area of Bangladesh	IID	n.o	Ma, Mi (LM)	Cultivated in open waters	Sarker et al. (2021)
Eucheuma denticulatum	Southwest coast of Madagascar	IID	High water temperature and low water nitrogen	Ma	Monoline	Mollion and Braud (1993)
	Northeastern Bohol, Philippines	IID	*Cytophaga-Flavobacterium* complex and *Vibrio* spp.	Ma, PI	Cultivated in open waters	Largo et al. (1995)
	Zanzibar, Tanzania	IID IID	High water temperature and light intensity Marine bacteria	Ma Ma	Off-bottom monoline Off-bottom	Largo et al. (2020) Ndawala et al. (2022)
	Tawi-Tawi, Philippines	IID	No significant association of temperature or salinity with disease prevalence	Ma	Floating and fixed-off bottom	Tahiluddin and Damsik (2023)
	Southern Kenya coast, Kenya	IID	Low water motion, high water temperature and salinity	Ma	Fixed off-bottom	Wakibia et al. (2006); Kimathi et al. (2018)
	Southeast Sulawesi, Indonesia	IID	*Vibrio*	Ma, PI	Horizontal net cage	Darma et al. (2021)
Gelidium	Coastal area of Bangladesh	IID	n.o	Ma, Mi (LM)	Cultivated in open waters	Sarker et al. (2021)
Gracilariopsis lemaneiformis	Shandong, China	RD	*V. alginolyticus*	Ma, Mi (SEM), Mo, PI	n.m	Pang et al. (2024, 2025)
Gracilaria verrucosa	South Sulawesi, Indonesia	IID	*Vibrio*, *Chromobacterium*, *Flavobacterium*, *Pseudomonas*, and *Achromobacter*; high water temperature	Ma, PI	Cultivated in pond	Zainuddin et al. (2019)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
Hypnea	Coastal area of Bangladesh	IID	n.o	Ma, Mi (LM)	Cultivated in open waters	Sarker et al. (2021)
Kappaphycus spp.	Tawi-Tawi, Philippines	IID	Marine bacteria	Ma	Fixed-off bottom	Tahiluddin and Damsik (2023)
	Zanzibar, Tanzania	IID	Marine bacteria	Ma	Off-bottom	Ndawala et al. (2022)
Kappaphycus alvarezii	Rariraki, Fiji	IID	High water temperature	Ma	Monoline	Luxton et al. (1987)
	Kochi, Japan	IID	Proliferation of bacteria	Ma, PI	Floating rafts	Largo et al. (1997)
	Hainan, China	IID	High water temperature and low salinity	Ma	Floating rafts	Pang et al. (2015); Pang and Liu (2019)
	Pangasinan, Philippines	IID	High light intensity, low water motion and nitrogen limitation	Ma	Bottom monoline	Trono and Valdestamon (1994)
	Northeastern Bohol, Philippines	IID	*Cytophaga-Flavobacterium* complex and *Vibrio* spp.	Ma, PI	Cultivated in open waters	Largo et al. (1995)
	Southern Luzon, Philippines	IID	*Aspergillus ochraceus*, *A. terreus*, and *Phoma* sp.	Ma, Mo, PI	Cultivated in open waters	Solis et al. (2010)
	Palawan, Philippines	IID	n.o	Ma	Floating bamboo raft	Plasus and Zabala (2020)
	Eastern Samar, Philippines	IID	n.o	Ma	Monoline	Yap-Dejeto et al. (2021)
	Tawi-Tawi, Philippines	IID	No significant disease correlation with temperature, salinity, pH, current, or wind speed	Ma	Cultivated in open waters (shallow and deep water)	Tahiluddin and Terzi (2024)
	Southern Kenya coast, Kenya	IID	Low water motion, high water temperature and salinity	Ma	Fixed off-bottom	Wakibia et al. (2006); Kimathi et al. (2018)
	South Sulawesi, Indonesia	IID	*Stenotrophomonas*, *Ochrobactrum*, and *Pseudomonas*	Ma, Mo, PI	Cultivated in open waters	Achmad et al. (2016)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
	South Sulawesi, Indonesia	IID	n.o	Ma	Cultivated in open waters	Rantetondok and Latama (2017)
	North Java Sea, Indonesia	IID	*Alteromonas*, *Bacillus*, *Pseudomonas*, *Pseudoalteromonas*, *Glaciecola*, *Aurantimonas*, and *Rhodococcus*	Ma, Mo	Cultivated in open waters	Syafitri et al. (2017)
	Central Sulawesi, Indonesia	IID	High water temperature	Ma	Longline	Ndobe et al. (2020)
	West Nusa Tenggara, Indonesia	IID	*Vibrio* spp.	Ma, PI	Longline	Setyowati and Cokrowati (2021)
	North Kalimantan, Indonesia	IID	*Vibrio*, *Pasteurella*, *Edwardsiella*, *Plesiomonas*, and *Chromobacterium*	Ma, PI	Cultivated in open waters	Azis et al. (2022)
	Maluku, Indonesia	IID	*V. alginolyticus*, *V. fluvalis*, *V. cholerae*, and *Aeromonas caviae*; high water temperature and salinity	Ma, PI	Cultivated in open waters	Tuhumury et al. (2024)
	Sabah, Malaysia	IID	Fluctuations in water salinity, temperature, and nutrient levels	Ma	Cultivated in open waters	Vairappan et al. 2008
	Sabah, Malaysia	IID	*Vibrio*, *Halomonas*, *Aestuariibacter*, and *Alteromonas*	Ma, Mi (LM)	Cultivated in open waters	Azizi et al. (2018)
	Sabah, Malaysia	IID	*Vibrio owensii* and *V. parahaemolyticus*	Ma, Mo	Cultivated in open waters	Mangun et al. (2023, 2025)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
	Southeastern, India	IID	High water temperature, low salinity and water movement, and increase in bacterial diversity	Ma	Monoline and floating rafts	Arasamuthu and Patterson (2018)
	Tamil Nadu, India	IID	Low water movement, high water temperature, and light intensity	Ma, PI	Cultivated in open waters	Riyaz et al. (2019)
	Gulf of Mannar, India	IID	Extreme changes environmental conditions	Ma, Mo	Cultivated in open waters	Riyaz et al. (2020)
	Tamil Nadu, India	IID	Marine bacteria	Ma, Mi (LM), Mo, PI	Cultivated in open waters	Riyaz et al. (2021)
	Zanzibar, Tanzania	IID	High water temperature and light intensity	Ma	Cultivated in open waters	Largo et al. (2020)
Kappaphycus striatus	Batangas, Philippines	IID	n.o	Ma	Cultivated in open waters	Mendoza et al. (2002)
	Southern Luzon, Philippines	IID	*Aspergillus ochraceus*, *A. terreus*, and *Phoma*	Ma, Mo	Cultivated in open waters	Solis et al. (2010)
	Palawan, Bohol, Zamboanga City, Davao del Norte, and Tawi-Tawi, Philippines	IID	Total monthly rainfall and low wind speed	Ma	Fixed-off bottom	Faisan et al. (2021)
	Tawi-Tawi, Philippines	IID	No significant disease correlation with temperature, salinity, pH, current, or wind speed	Ma	Cultivated in open waters (shallow and deep)	Tahiluddin and Terzi (2024)
	Zanzibar, Tanzania	IID	High water temperature and light intensity	Ma	Cultivated in open waters	Largo et al. (2020)
	Khanh Hoa, Vietnam	IID	*Vibrio*, *Cobetia*, *Marinomonas*, and *Pseudoalteromonas*	Ma, Mo	Cultivated in open waters	Kopprio et al. (2021a)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
Palmaria palmata	Scotland, UK	OD	*Olpidiopsis palmariae*	Ma, Mi (LM), Mo	Cultivation lines seeded and culture tanks	Badis et al. (2019)
Pyropia	Chiba, Japan	RRD	*Pythium porphyrae*	Ma, Mi (LM), PI	n.m	Uppalapati and Fujita (2000)
	Seochon, South Korea	OD	*Olpidiopsis*	Ma, Mi (LM, TEM), PI	n.m	Kim et al. (2014)
		GSD	*Flavobacterium*, *Pseudoalteromonas*, and *Vibrio*	Ma, Mi (LM, TEM), PI	n.m	
		RRD	*P. porphyrae*	Ma, Mi (LM), Mo, PI	n.m	Badis et al. (2020a)
Pyropia haitanensis	Jiangsu, China	YSD	*Vibrio mediterranei*	Ma, Mi (LM, TEM), Mo, PI	Nursery farm	Yang et al. (2020)
	Shantou, China	GSD	Marine bacteria	Ma, Mi (LM),	Floating rafts	Bao et al. (2025)
	Jiangsu, China	RSD	*Vibrio fortis*; high water temperature	Ma, Mi (LM, SEM), Mo, PI	n.m	Luo et al. (2025)
Pyropia yezoensis	Chiba, Japan	AD	*Flavobacterium*	Ma, Mi, PI	n.m	Sunairi et al. (1995)
	South Jeolla, South Korea	RRD	*P. porphyrae*	Ma, Mi (LM), Mo, PI	Cultivated in open waters	Park et al. (2000); Hwang et al. (2009)
	South Jeolla, South Korea	RRD	*P. porphyrae*	Ma, Mi (LM, TEM), PI	Cultivation nets	Kim et al. (2014)
	Seocheon, South Korea	OD	*Olpidiopsis pyropiae*	Ma, Mi (LM, TEM), Mo, PI	n.m	Klochkova et al. (2016)
	Jindo and Seocheon, South Korea	GSD	*Pyropia*-infecting virus 1	Ma, Mi (LM, TEM), PI	n.m	Kim et al. (2016)
	North Jeolla, South Korea	RRD	*Pythium chondricola* and *P. porphyrae*	Ma, Mi (LM), Mo	n.m	Lee et al. (2017)
	Western Sea of Korea	OD	*O. pyropiae*	Ma, Mi (LM, TEM), Mo, PI	n.m	Kwak et al. (2017)
	Jindo, South Korea	RRD	*P. porphyrae*	Ma, Mi (LM), PI	Cultivated in open waters	Kim et al. (2023)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
	Seocheon, South Korea	OD	*O. pyropiae*	Ma, Mi (LM), PI	Cultivated in open waters	Kim et al. (2023)
	Incheon, South Korea	DS	*Pythium*, *Olleya*, *Aspidisca*, and *Zosterodasys*	Ma, Mi (LM), Mo	Cultivated in open waters	Bae et al. (2024)
	Jiangsu, China	RRD	*P. chondricola*	Ma, Mi (LM), Mo	Cultivated in open waters	Yan et al. (2019)
	Jiangsu, China	WSD	*Phoma*; high water temperature	Ma, Mi (LM, SEM), Mo, PI	Breeding farm	Guan et al. (2013)
	Jiangsu, China	RRD	*P. chondricola*; high water temperature and salinity	Ma, Mi (LM), Mo, PI	Half-floating culture	Qiu et al. (2019)
	Jiangsu, China	YSD	Bacteria and several specific pathogens	Ma, Mo	Conchocelis sporeling ponds	Liu et al. (2020)
	Jiangsu, China	RRD	*Alternaria*; high water temperature and low salinity	Ma, Mi (LM, SEM), Mo, PI	Floating rafts	Mo et al. (2016)
	Jiangsu, China	RRD	*P. porphyrae*; high water temperature, low salinity, and heavy rain	Ma, Mo	Floating rafts	Ding and Ma (2005)
	Haizhou Bay, China	RRD	*Pythium*; high water temperature and low salinity	Ma, Mo	Cultivated in open waters	Liu et al. (2024a)
	Jiangsu, China	BD	*Phaeobacter italicus* JN-W1; high water temperature and pH	Ma, Mi (LM, TEM), Mo, PI	n.m	Liu et al. (2025b)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
Saccharina japonica	Hokkaido, Japan	RRD	*Alteromonas*; high water temperature	Ma, IP	Nursery tank	Yumoto et al. (1989)
	Shandong, China	HRD	*Pseudoalteromonas*, *Vibrio*, and *Halomonas*	Ma, Mi (TEM), Mo, PI	n.m	Wang et al. (2008)
	Shandong, China	HRD	*Sulfitobacter* and *Loktanella*	Ma, Mo	n.m	Zhang et al. (2020)
	Fujian, China	GRD	*Exiguobacterium oxidotolerans* XP-2; high water temperature and light intensity	Ma, Mi (LM, SEM, TEM), Mo, PI	n.m	Ahmad et al. (2021)
	Shandong, China	WSD	Marine bacteria; windless, high water temperature, and light intensity	Ma, Mi (LM)	Cultivated in open waters (shallow)	Wang et al. (2021)
	Shandong, China	GRD	*Gammaproteobacteria*, *Alphaproteobacteria*, and *Bacteroidetes*; low light intensity and incorrect nutrient	Ma, Mi (LM), Mo	Seedling hatchery	Yan et al. (2022)
	Shandong, China	BD	*Pseudoalteromonas piscicida* X-8	Ma, Mi (LM, SEM, TEM), Mo, PI	Nursery farm	Zhang et al. (2022)
	Shandong, China	WRD	Marine bacteria; low nitrogen and phosphorus	Ma, Mo	Floating rafts	Ma et al. (2024)
Saccharina latissima	British Columbia, Canada	PSD	*Algicola*	Ma, Mo	Nursery farm	Schenk et al. (2025)
	South and west, Netherlands	BD	Phaeovirus and ormycovirus	Ma, Mo	n.m	Dekker et al. (2024)

Table 1 (continued)

Host seaweed	Location/country	Diseases	Causative agents	Diagnostic methods	Farming method	References
Sargassum	Coastal area of Bangladesh	IID	n.o	Ma	Cultivated in open waters	Sarker et al. (2021)
Ulva australis	Zeeland, Netherlands	BD	Picorna-like viruses may be involved	Ma, Mo	Cultivated in ponds	van der Loos et al. (2023)
Undaria pinnatifida	Southeastern and Southwestern of Korea	PD	*Amenophia orientalis* and *Scutellidium*	Ma, Mi	Pole cultivation	Park et al. (1990)

Diseases: see Table 2 for the full forms of all disease abbreviations in this column (Table 2 is a continuation of Table 1). Causative agents: *n.o.*, not observed; *n.m.*, not mentioned. Diagnostic methods: *Ma*, macroscopic; *Mi*, microscopic (*LM*, light microscope; *SEM*, scanning electron microscopy; *TEM*, transmission electron microscopy); *Mo*, molecular; *PI*, pathogen isolation

Table 2 Disease symptoms across diverse farmed seaweed species, reported in both open-water culture and hatchery facilities

Disease name	Host	Symptoms
Anaaki disease (AD)	*P. yezoensis*	Small perforations first develop at the centre of the thallus and gradually enlarge, eventually causing the thallus to detach from the culture base and be carried away by tides and wind. Lesions are predominantly central rather than marginal and are not accompanied by discolouration
Bleaching disease (BD)	*S. japonica*, *S. latissima, P. yezoensis*, and *U. australis*	The emergence of distinctly demarcated white spots, streaks, or patches on the seaweed blade; however, tissues beyond the affected area remain healthy and continue normal growth
Disease spot (DS)	*P. yezoensis*	Appearance of speckled areas on the thallus with discolouration to white, pink, or green. In some cases, the lesions progress to involve the full thickness of the blade tissue
Green rot disease (GRD)	*S. japonica*	Initiated by loss of brown pigment at the seedling tips, progressing to gradual rot that ultimately destroys the entire seaweed blade
Green spot disease (GSD)	*Pyropia* spp.	A green chain of dead, coalescing cells forms at the thallus wound, followed by the gradual decay of the entire seaweed blade
Hole rotten disease (HRD)	*S. japonica*	The appearance of small perforations on the blade that coalesce, progressively enlarge, and ultimately destroys the entire seaweed blade
Ice-ice disease (IID)	*C. racemosa*, *Enteromorpha*, *E. denticulatum*, *Gelidium*, *G. verrucosa*, *Hypnea*, *Sargassum*, and *Kappaphycus* spp.	The thallus becomes whitened, translucent, and mucilaginous to the touch. This physical degradation leads to biomass loss and a marked decline in carrageenan content
Olpidiopsis disease (OD)	*Pyropia* spp. and *P. palmata*	Starting with blade bleaching that progresses to greenish lesions, this disease causes rapid tissue decay culminating in total mortality of the host
Pinhole disease (PD)	*U. pinnatifida*	The presence of galls manifesting as small perforations (0.5–1.5 mm) on the blades, stipes, and sporophylls; seaweed cells at the perforation margins exhibit markedly stronger brown pigmentation than normal cells in uninfested thalli
Pink spot disease (PSD)	*S. latissima*	The emergence of small, flat, pink patches on the surface of seaweed filaments. Over time, these patches spread laterally across the surface, resulting in persistent pigmentation that is extremely difficult to remove from the spools
Red rot disease (RRD)	*Pyropia* spp. and *S. japonica*	Small red patches that spread, accompanied by numerous perforations, ultimately causing the entire blade to disintegrate
Rot disease (RD)	*G. lemaneiformis*	Discolouration of the thallus from red to pink and ultimately white, with the most severe decay at the basal region proximal to the attachment line

Table 2 (continued)

Disease name	Host	Symptoms
Rough-shell disease (RSD)	*P. haitanensis*	The appearance of white patches, accompanied by roughening of the seaweed thallus surface
White rot disease (WRD)	*S. japonica*	Initially, lesions appear at the actively growing thallus tips; these then spread centrally, arresting seaweed growth. Over time, the infected tissue ulcerates or decays and ultimately sloughs off from the main blade
White spot disease (WSD)	*P. yezoensis* and *S. japonica*	Distinct, irregular white patches are present, sometimes accompanied by very large blisters exceeding 10 cm. These lesions occur on the upper and central regions of the sporophyte
Yellow spot disease (YSD)	*P. haitanensis* and *P. yezoensis*	Early symptoms appear as yellow spots along the thallus margin, accompanied by lesions; these then spread and turn white, ultimately resulting in complete bleaching of the seaweed

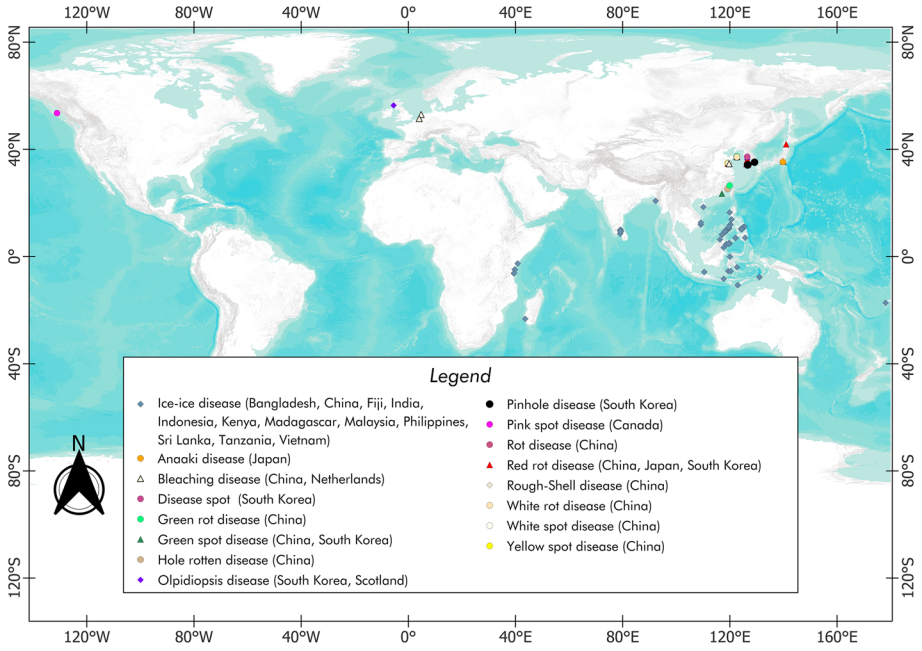


Fig. 1 Global distribution of reported seaweed diseases in aquaculture

By 2025, diseases in seaweed aquaculture have been reported across multiple regions (see Fig. 1) and constitute a principal driver of the global decline in production (Sugumaran et al. 2022). Notable outbreaks were documented in Africa, including Tanzania in 2020 (Kunzmann et al. 2023). In Asia, analogous events were reported in China during 2017–2018 (Wang et al. 2021) and in 2023 (Pang et al. 2024), as well as in several production hubs in the Philippines (Faisan et al. 2021; Tahiluddin and Terzi 2024). Beyond yield losses, the spread of pathogenic spores threatens the long-term viability of cultivation (Zhang et al. 2022). Among eucheumatoids, carrageenan extracted from diseased thalli exhibits inferior quality compared to that from healthy seaweeds (Riyaz et al. 2022). The economic consequences are likewise substantial, as recorded in Tanzania in 2020 (Kunzmann et al. 2023). Given this urgency, a deeper understanding of the pathogens and diseases affecting seaweed aquaculture is imperative. Ward et al. (2020) provided a comprehensive review, yet the analysis was oriented toward Asia, while seaweed production occurs across diverse regions globally. Accordingly, this review seeks to broaden the scope by mapping recent advances in understanding seaweed diseases and their associated biotic and abiotic drivers and evaluating the effectiveness of proposed solutions to date. To this end, we conducted an exhaustive appraisal of literature indexed in Scopus, Web of Science, and SpringerLink, complemented by additional sources from Google Scholar and FAO, published up to July 2025. This review aims to deliver robust scientific insights into disease threats in seaweed aquaculture and to promote healthier and more sustainable production practices.

Table 3 Search terms used to identify articles for review

Keywords	
	(disease* OR infect*) AND (seaweed* OR macroalga*)
AND	(cultivat* OR farm* OR aquacultur* OR maricultur*)
AND	(pathogen* OR bacteri* OR vir* OR fung* OR parasite*)
OR	("environmental condition*" OR "environmental factor*" OR "water quality" OR "temperature stress" OR "salinity stress" OR "nutrient stress" OR "climate change")

* indicates that any word following it may be used as a search term

Publication screening

This systematic review was reported in accordance with the PRISMA guidelines as articulated by Charlton et al. (2016) for reporting systematic literature reviews. Literature searches were conducted in three core databases: Scopus, Web of Science, and SpringerLink, and supplemented by searches via Google Scholar and online resources including the website of the FAO using the same keyword strategy. The search strategy employed Boolean operators, and search settings are detailed in Table 3. The temporal scope was limited to peer-reviewed articles published in English up to July 2025.

Inclusion criteria required publications to be reviews, reports, or research articles that documented disease occurrence in seaweed populations at nursery units or during active farming. A case was deemed eligible when (i) diseased seaweeds originated from nursery or farming settings, (ii) the disease name was stated, (iii) the location of occurrence was clearly reported, (iv) symptoms were recorded during the nursery or farming phase, and (v) at least one form of verification was provided through macroscopic, microscopic, or molecular assessment supported by photographic documentation and/or symptom descriptions. Laboratory follow-up studies were retained when plant material originated from nurseries or farms currently or previously affected by disease rather than from experimental induction. Publications were excluded if they were reviews lacking specification of the disease name, the infected seaweed species, the identification method, and a clearly defined location, or if diseased material derived from controlled experiments with induced disease that did not reflect field conditions.

Given the breadth of search terms required to identify articles on disease occurrence, the initial database queries yielded a large number of records. Irrelevant subject areas were excluded using database filtering functions, as summarised in Table 4.

All search records were exported to EndNote X7 for systematic deduplication. Study selection proceeded with title and abstract screening to apply the inclusion–exclusion criteria consistently, followed by full-text eligibility assessment. Two reviewers (A.W.F.A. and K.K.) conducted these steps independently to reduce selection bias, with disagreements resolved through discussion until consensus was reached. In total, 75 articles met the criteria and were included in the final synthesis, as summarised in Fig. 2. Data extraction was performed in a structured manner for each article, capturing the cultivated seaweed species, farming location, reported causal factors (abiotic and biotic), approaches to disease identification, and farming methods. The extracted data are presented in Tables 1 and 2, and a spatial map of disease distribution across farming sites is shown in Fig. 1. These procedures ensure that the search, selection, and synthesis align with best practice in systematic reviews and provide a robust foundation for subsequent thematic analyses.

Table 4 Subject areas excluded using database filtering

Scopus	WoS	SpringerLink
Veterinary	Zoology	Plant Materials
Social Sciences	Water Resources	Organic Farming
Pharmacology, Toxicology and Pharmaceutics	Veterinary Sciences	Marine Biology
Nursing	Toxicology	Limnology
Neuroscience	Soil Science	Freshwater and Marine Ecology
Materials Science	Public Environmental Occupational Health	Biofuels
Health Professions	Polymer Science	
Engineering	Physiology	
Energy	Physics Atomic Molecular Chemical	
Computer Science	Pharmacology Pharmacy	
Chemistry	Parasitology	
Chemical Engineering	Oncology	
Business, Management, and Accounting	Nutrition Dietetics	
	Mycology	
	Medicine Research Experimental	
	International Relations	
	Integrative Complementary Medicine	
	Infectious Diseases	
	Immunology	
	Horticulture	
	Green Sustainable Science Technology	
	Genetics Heredity	
	Gastroenterology Hepatology	
	Forestry	
	Evolutionary Biology	
	Environmental Studies	
	Environmental Sciences	
	Engineering Multidisciplinary	
	Engineering Environmental	
	Engineering Chemical	
	Energy Fuels	
	Endocrinology Metabolism	
	Ecology	
	Chemistry Physical	
	Chemistry Organic	
	Chemistry Multidisciplinary	
	Chemistry Medicinal	
	Chemistry Inorganic Nuclear	
	Chemistry Applied	
	Biochemistry Molecular Biology	
	Agronomy	
	Agriculture Multidisciplinary	

Table 4 (continued)

Scopus	WoS	SpringerLink
	Agriculture Dairy Animal Science Agricultural Engineering	

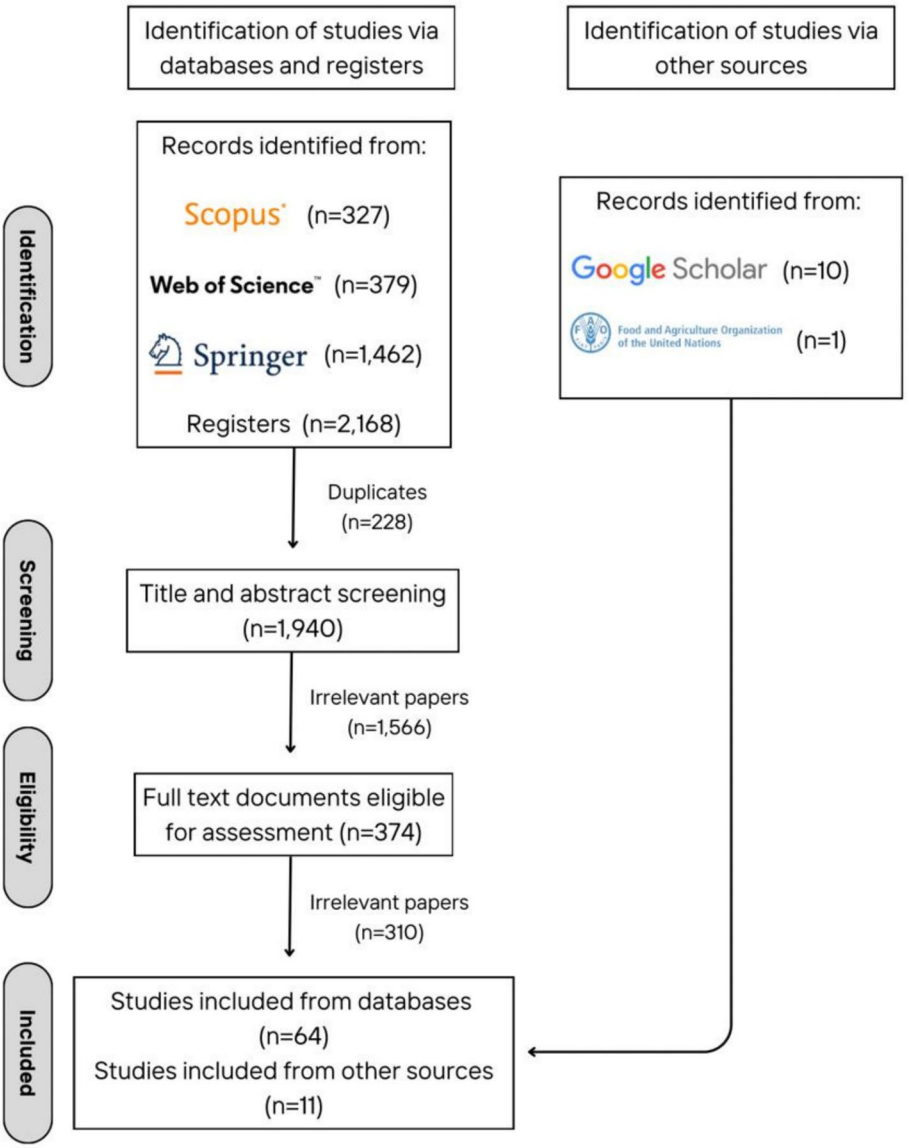


Fig. 2 PRISMA flow diagram used to retrieve and filter papers related to the review

Biotic and abiotic drivers of disease in cultivated seaweeds

Numerous reports of diseases in cultivated seaweed have occurred across multiple regions (see Table 1), in parallel with the rapid expansion of aquaculture and with variability in the marine environment, including fluctuations in temperature, salinity, acidification, nutrient availability, light intensity, and current dynamics, which together have amplified the frequency and spatial extent of pathogen outbreaks (Ren et al. 2022; Li et al. 2023; Zhang et al. 2024b). From a biotic perspective, numerous studies have documented the involvement of bacteria, parasites, and viruses across commercial seaweed species (Friedlander and Critchley 2024), whereas abiotic factors modulate host susceptibility through physiological stress and shifts in the microbiome on the surface as well as within tissues (Ghaderiardakani et al. 2020; Martínez et al. 2024). Within this framework, a microbial invasion is considered pathogenic when effective contact with the host triggers abnormal physiological processes that disrupt morphology, growth, performance, and reproductive success of seaweeds, either through direct interactions or tissue penetration, accompanied by variable host defence responses (Hurtado et al. 2019; Lang et al. 2024).

Biotic drivers

Bacteria

Associations between bacteria and seaweeds are ubiquitous: bacteria occur as epibionts on the thallus or as endobionts within tissues, often assembling into surface biofilms, and together with the host they constitute a holobiont characterised by chemical signalling and nutrient exchange (Lu et al. 2023; Saha et al. 2024; Umashree et al. 2024). Bacterial presence on seaweeds is highly diverse and is typically shaped by host species and environmental conditions (Kaur et al. 2023). Nevertheless, numerous studies show that *Proteobacteria* and *Bacteroidetes* dominate the microbiomes of red seaweeds (Xie et al. 2024), brown seaweeds (Ihua et al. 2020), and green seaweeds (Califano et al. 2020). Community composition is dynamic: environmental drivers such as temperature, nutrients, light, and season modulate the relative abundance of bacterial groups (Lu et al. 2023). In addition, the diversity of bacterial communities is influenced by seaweed exudates (for example monosaccharides, amino acids, and polysaccharides), which play a key role in structuring communities and shaping gene-expression patterns. Epiphytic bacteria are enriched for genes that degrade algal exudates and host cell-wall polysaccharides, including algL (alginate lyase) and agarase (Pfister et al. 2025). Chemical interactions between seaweed exudates and bacteria induce the expression of substrate-specific transport and metabolic genes in bacteria (Ahmed et al. 2021). In summary, bacteria exploit seaweed-derived nutrients as sources of carbon and energy, activating genetic pathways for the uptake and degradation of host-secreted organic compounds (Pfister et al. 2025).

These bacteria can contribute to the seaweed life cycle by producing regulatory compounds and secondary metabolites that support growth, adaptation, morphogenesis, and defence (Hmani et al. 2023; Lang et al. 2024). For example, several epiphytic isolates produce enzymes such as amylase, lipase, and agarase that facilitate the degradation of organic matter, as well as antimicrobial compounds that protect seaweed against pathogens (Pinteus et al. 2021; Hmani et al. 2023). Bacterial species from the genus

Paracoccus attached to *Ulva* are known to release regulatory molecules that stimulate seaweed growth and morphogenesis (Hmani et al. 2023). Complementing this, functional epiphytic bacteria on *U. pinnatifida* can enhance key host physiological processes by augmenting carbon and nitrogen assimilation (Chen et al. 2025). Thus, companion bacteria underpin seaweed life cycles by supplying essential nutrients, providing molecular signals for environmental adaptation, and producing growth-promoting regulators. In return, bacteria obtain nutrition from seaweed exudates and metabolic residues, consistent with the enrichment of amino-acid and carbon transport genes in host-associated bacteria (Pfister et al. 2025).

However, this symbiosis is not invariably mutualistic; under certain conditions, epiphytic bacteria can become opportunists or pathogens (Saha and Weinberger 2019). In *P. yezoensis*, for instance, harmful bacterial consortia proliferate at elevated temperatures and can cause tissue degradation (Zeb et al. 2024). Thick biofilms on the thallus surface can block light penetration required for photosynthesis, reduce nutrient availability, and hinder gas exchange, thereby markedly impairing seaweed growth (Kaur et al. 2023; Senthil Rathi et al. 2024). On contaminated thalli, biofilm formation is often dominated by *Vibrio* spp. and *Bacillus* spp., which readily colonise the organic surface matrix (Pinteus et al. 2021). Seaweed responses to biofilms are not uniform; some species deploy chemical and physical defences that inhibit initial colonisation and biofilm maturation. For example, the *Halymenia* releases allelopathic compounds that disrupt biofilm development and reduce microbial attachment to the thallus surface (Abdul Malik et al. 2020). By contrast, *Kappaphycus* spp. frequently experience biofilm attachment on the thallus. Although such biofilms do not always trigger disease directly, their presence correlates with increased microbial growth that can heighten host susceptibility (Kambey et al. 2021b). Consequently, biofilm control is an operational priority in aquaculture systems, and routine cleaning of thalli and culture substrates has been reported to effectively reduce biofilm accumulation and, in turn, the risk of seaweed health disorders (Kambey et al. 2021b).

The composition of bacterial communities adhering to seaweeds is strongly conditioned by the environment, and pathogenic potential can shift accordingly (see Fig. 3). Bacteria that are common in healthy individuals may become opportunistic pathogens when conditions change. For example, bacteria and cyanobacteria that are native members of the epiphytic microbiome on *Pyropia* spp. blades can turn opportunistic under certain environmental regimes, such as increased seawater temperature (Kim et al. 2014; Zeb et al. 2024). *Exiguobacterium oxidotolerans* XP-2, initially dominant on healthy *S. japonica* thalli, was identified as an opportunistic pathogen that caused seaweed mortality in laboratory assays, as demonstrated by fulfilment of Koch's postulates (Ahmad et al. 2021). Similarly, *Pseudomonas* isolates that occur naturally on healthy *S. japonica* thalli can induce disease symptoms under controlled conditions, indicating that pathogenic behaviour emerges when environmental factors and host status are permissive (Yang et al. 2023). Collectively, these findings underscore a context-dependent boundary between bacterial commensalism and pathogenicity in seaweeds and highlight the importance of farm-management practices that prevent microbiome shifts toward disease-promoting states. Bacteria–seaweed associations also exhibit marked host specificity. For instance, a particular strain of *V. alginolyticus* (X-2) that naturally colonises *S. japonica* confers protection by reducing the risk of BD in that seaweed (Ma et al. 2023). In contrast, other *V. alginolyticus* strains have been reported as pathogens causing RD (locally named "Baotou") in the agarophyte *G. lemaneiformis* (Pang et al. 2024, 2025). These contrasting cases illustrate how environmental change and host identity can shift bacterial roles from mutualistic to pathogenic, indicating that pathogenicity risk should be assessed at the strain level rather than solely at the species level.

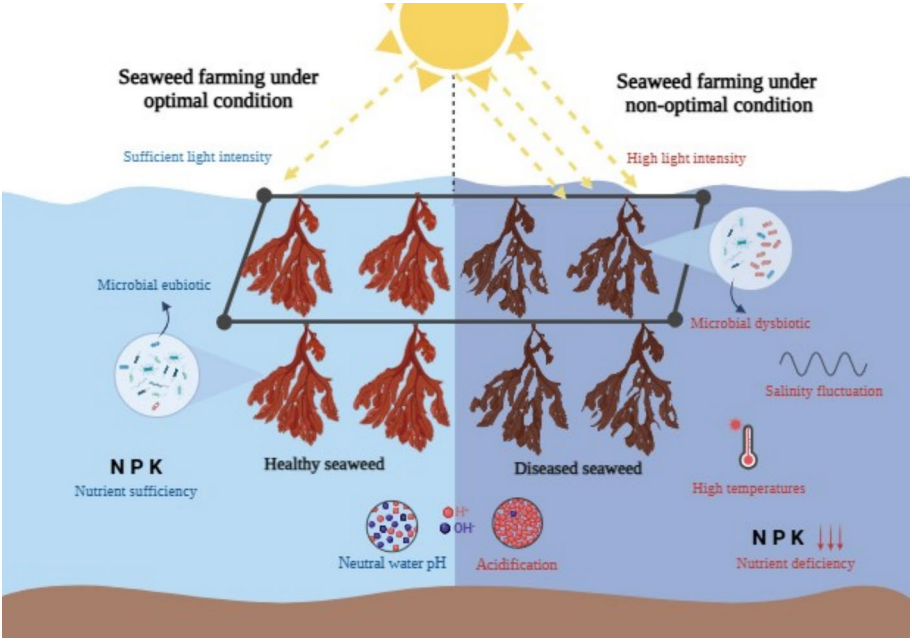


Fig. 3 Schematic comparison of seaweed aquaculture under optimal versus non-optimal conditions: under optimal conditions, a microbial eubiotic is maintained and seaweeds remain healthy; in contrast, multi-stressor pressures (high light intensity, ocean warming, acidification, salinity fluctuation, and nutrient deficiency) induce microbial dysbiosis, heighten pathogenic infection risk, and ultimately result in diseased seaweed

Parasites

Parasitism is an interaction in which a parasite benefits at the expense of its host, in this case cultivated seaweeds. At the cellular level, parasitic colonisation is often accompanied by the release of enzymes that degrade the cell wall and exploit structural polysaccharides as nutrient sources, ultimately undermining tissue integrity and the seaweed's metabolic capacity (Jormalainen et al. 2001). As of 2025, accumulating empirical evidence indicates that parasites are key determinants of seaweed health (Table 1) and major drivers of productivity decline across several seaweed commodities, particularly in East Asia. Among the most consistently reported groups are the oomycetes, with rising incidence in recent years across multiple production hubs (Zuccarello et al. 2024; Friedlander and Critchley 2025b). *Olpidiopsis pyropiae* has been identified as a principal pathogen in *Pyropia* aquaculture in South Korea, causing spots that damage thallus blades and degrade the quality of both dried and fresh products (Kim et al. 2023; Bae et al. 2024). Beyond *Olpidiopsis*, *Pythium* species such as *P. porphyrae* and *P. chondricola* also infect *Pyropia* and contribute to outbreaks, including reports of RRD and RSD that can precipitate harvest failures from local to regional scales (Badis et al. 2020a; Lee and Lee 2022; Bae et al. 2024). Although taxonomically related, *Olpidiopsis* and *Pythium* exhibit contrasting infection biology with direct management implications. *Olpidiopsis* is an obligate parasite that infects only living cells; its cycle involves the release of zoospores from zoosporangia, and these zoospores remain viable in the

environment for approximately 1 week, during which they must locate a new host to continue the cycle. By contrast, *Pythium* can persist as a saprotroph on decaying seaweed tissue during the off-season, maintaining pathogen reservoirs even when primary host populations decline (Park et al. 2025). These differences help explain variability in field outbreak dynamics and necessitate tailored interventions. Oomycete infections broadly disrupt tissue architecture and function, inducing morphological change, physiological impairment, and reduced growth rates that culminate in losses of yield and quality, with tangible economic impacts for farmers (Spagnuolo and Genovese 2024; Kim et al. 2023).

An ecological perspective further clarifies that risk does not originate solely within farmed systems. Recent observations show that parasites and gall formation are also common in wild seaweed populations, indicating that infected natural populations may act as reservoirs or vectors transmitting pathogens to farms via currents, thallus fragments, or biotic intermediaries. Reports from Chile and the Southeast Pacific document the first cases of *P. porphyrae* infection in *Pyropia* in that region. At *Gracilaria chilensis* farms where *Pyropia* is often considered a fouling pest on culture infrastructure, outbreaks of *P. porphyrae* were followed by the collapse of *Pyropia* populations within weeks, suggesting that pathogens can function as natural pest regulators in *Gracilaria* polyculture, albeit with residual risk across commodities (Munoz et al. 2024). In South Korea, *Bangia fuscopurpurea*, commonly viewed as a nuisance on *Pyropia* culture nets, serves as an alternative host for *P. porphyrae* and *Olpidiopsis*, thereby expanding the infection range and prolonging pathogen life cycles in farm environments. In *P. yezoensis*, infections originating in *Bangia* can spread to newly developing *Pyropia* thalli, forming an epidemiological bridge that heightens outbreak risk if unmanaged (Park et al. 2025). Collectively, these findings underscore the need to map interspecific interaction networks within and around farming sites, because species once regarded as mere nuisances may in fact operate as crucial links in pathogen transmission.

Viruses

The presence and dynamics of viruses in seaweed aquaculture are governed by interactions between abiotic stressors and host susceptibility. Water pollution and other anthropogenic pressures compromise seaweed physiological health and may facilitate viral infection, while physical characteristics of the environment such as water movement and wave exposure influence infection prevalence in particular species (Lovdal et al. 2021; Martínez et al. 2024; Fig. 3). Climate warming is expected to increase the incidence of viral disease, consistent with the role of viruses as major biotic agents of mortality for many marine organisms, including seaweed (Benites and Alves-Lima 2022; Zhao et al. 2024). To date, viral invasion has been reported in at least 54 seaweed species (Schroeder and Mckeown 2021). The diversity of seaweed-infecting viruses can be grouped broadly into non-pathogenic or even host-beneficial viruses, potential pathogens that may cause disease under specific conditions, and virulent pathogens that pose substantial risks to seaweed aquaculture and global food security through damage to hatcheries and farms (Mizutani et al. 2022; Murúa et al. 2023; Benites and Alves-Lima 2022). Despite growing inventories, our understanding of the full virosphere associated with seaweeds remains limited, research on red seaweed viruses has progressed more slowly than work on green seaweed viruses, and only a handful of seaweed viruses have

been comprehensively characterised (Benites and Alves-Lima 2022; Martínez et al. 2024).

Recent molecular evidence is beginning to resolve more nuanced causal relationships among viruses, host status, and environmental context. Virome surveys and molecular assays have identified numerous previously unrecognised viruses in seaweeds. For example, in the green seaweed *Ulva*, several Circular rep-encoding single-stranded (CRESS) DNA viruses and picorna-like viruses were highly abundant on bleached thalli but absent or rare on healthy thalli (van der Loos et al. 2023). In the brown seaweed *S. latissima*, phaeoviruses with double-stranded DNA were detected within host tissues (Dekker et al. 2024). In *A. esculenta*, both phaeoviruses and an unclassified RNA virus designated Aev-NL1, related to ormycoviruses, were associated with the host (Dekker et al. 2024). Notably, asymptomatic *A. esculenta* also harboured these viruses, indicating that viral presence does not necessarily translate into disease phenotypes (Dekker et al. 2024). In *Ulva*, bleaching appears to involve both bacteria and viruses, including picorna-like and CRESS-DNA groups that proliferate in weakened or bacterially infected tissues, suggesting that co-infection may exacerbate disease progression. However, definitive causality still requires isolation of pure viral cultures and controlled infection trials (van der Loos et al. 2023). More direct aetiological evidence exists for *Pyropia*, in which a *Pyropia*-infecting virus has been linked to GSD in cultivation systems, demonstrating that specific viruses can act as primary agents under conditions conducive to replication and transmission in seaweed hosts (Kim et al. 2016). These observations underscore the need for further research focused on viral isolation, experimental demonstration of infectivity in seaweeds, and the exploration of interactions between viruses and co-occurring pathogens such as bacteria, together with their integrated impacts on hosts.

Importantly, not all seaweed-associated viruses are detrimental; some may be neutral or beneficial. Recent work detected RNA viruses in *S. latissima* across both bleached and apparently healthy individuals, with Slv-NL2 and Slv-NL3 present in all samples from disease-affected sites regardless of bleaching status, consistent with latent or asymptomatic infection (Dekker et al. 2024). Conversely, healthy *A. esculenta* contained dsDNA phaeovirus *Feldmannia irregularis* virus and several RNA viruses that were absent from diseased samples, implying that certain viruses can persist without causing pathology and may even associate passively with their hosts (Dekker et al. 2024). In red seaweed, viruses may adopt latent or lysogenic strategies, persisting across multiple host generations without overt cellular damage, in contrast to lytic infections (Benites and Alves-Lima 2022). Some persistent seaweed viruses are hypothesised to be mutualistic; for instance, members of the family Mitoviridae (ssRNA) occur persistently in cultured *P. yezoensis* and appear non-parasitic, with potential benefits to the host, although mechanisms remain unresolved (Mizutani et al. 2022). Such findings raise the possibility that certain viruses function analogously to beneficial endosymbionts, for example by stabilising the microbiome or enhancing host stress tolerance (Mizutani et al. 2022). Recent advances highlight seaweed virology as an increasingly vital field in aquaculture. Exploring viruses both as pathogens and as potential allies requires a holistic research agenda. Such efforts are expected to yield more effective disease control strategies and open opportunities to harness viruses for enhancing resilience and productivity in cultivation (Mizutani et al. 2022).

Abiotic factors

Abiotic factors such as temperature, salinity, light intensity, nutrient availability, water mass movement, and water quality modulate host physiological responses, alter the composition of the microbiome on seaweed surfaces and tissues, and ultimately increase susceptibility to opportunistic infection (Menaa et al. 2020; Veenhof et al. 2024; Zhang et al. 2024b). When environmental parameters exceed a species' tolerance range, seaweeds experience physiological stress marked by reduced photosynthetic rates, biomass loss, weakening of cell wall integrity, and disrupted metabolism, thereby elevating the likelihood of pathogen invasion (Hayashi et al. 2010; Tahiluddin and Terzi 2021; Martínez et al. 2024). These conditions align with findings that seaweed diseases tend to emerge and spread more rapidly under extreme environments, demonstrating a strong causal relationship between physicochemical anomalies of seawater and disease dynamics (Veenhof et al. 2024). In parallel, Caley et al. (2025) show that warming increases community variability and elevates the relative abundance of several putative pathogens. This relationship is also illustrated in Table 1, which summarises information on the role of various abiotic factors in triggering and exacerbating disease outbreaks. Likewise, Fig. 3 illustrates the differences in abiotic conditions in relation to seaweed conditions.

A close linkage between mechanism and field evidence is evident in *Pyropia* farming in South Korea. Planting calendars have traditionally been arranged to exploit cold temperatures from October to March in order to suppress RRD. The pathogen responsible for RRD, *P. porphyrae*, releases zoospores with higher success when water temperature rises and salinity declines, so winter months act as a natural brake on this disease cycle (Schroeder et al. 2013; Kim et al. 2023). Progressive winter warming and greater unpredictability have disrupted this brake, leading farmers not only to move to deeper waters but also to apply clean freezing at around − 20 °C before the planting season. This strategy reduces risk, yet it is energy and cost intensive and relies on synthetic materials that may increase pollution burdens if waste is not properly managed (Kim et al. 2023; Sugumaran et al. 2022). In China, *Pyropia* farmers have explored cultivation in deeper offshore waters as an adaptive strategy to improve thermal stability and mitigate disease risks, including seedling rot (Liu et al. 2025c). This approach underscores the importance of spatial planning as an environmental adaptation instrument (Wang et al. 2025). Thermal sensitivity has also been documented for *S. japonica* in southern China, specifically in Fujian Province (a subtropical region), where more than half of *S. japonica* production takes place. Rising temperatures have been reported to reduce seed quality, increase disease incidence, and under extreme conditions even prevent completion of the life cycle without human intervention during summer months (Zhang et al. 2020; Hu et al. 2021; Cottier-Cook et al. 2022). A body of evidence from cellular to field scales consistently indicates that abiotic variability is a primary driver of seaweed disease risk.

Special attention should be directed to IID, given its high incidence reported globally, particularly in seaweed aquaculture within tropical zones (Fig. 1). In intertidal farming systems in shallow waters, tidal dynamics induce direct exposure to solar radiation and fluctuations in temperature and salinity. Combined thermal and osmotic stress leads to depigmentation and tissue dysfunction in seaweed, followed by colonisation by opportunistic microorganisms on seaweed surfaces and thalli (McHugh 2003). In addition, elevated temperatures reduce the seaweed chemical defence capacity, increasing the likelihood of dysbiosis (Kumar et al. 2020; Kopprio et al. 2021b; Tahiluddin and Terzi 2021; Hudson et al. 2022; Rupert et al. 2023). Empirically, warmer waters are associated with higher

prevalence of IID in eucheumatoids and gracilarioids, reflected in reduced yields and carrageenan quality (Munisamy et al. 2024; Friedlander and Critchley 2025a). In Zanzibar, IID prevalence peaks in the hot dry season and declines markedly during the rainy season, indicating a strong linkage with seasonal thermal dynamics (Largo et al. 2020). Field reports from East Africa note sea surface temperatures exceeding 38 °C that trigger mass mortality in farmed seaweeds, while in India, recurrent marine heatwaves have been associated with IID outbreaks in *K. alvarezii* that disrupt the national carrageenan economy (Ndawala et al. 2021; Mantri et al. 2024; Dineshkumar et al. 2024; Innocentini et al. 2024). These data are consistent with the view that thermal anomalies and salinity fluctuations are important contributing factors associated with IID occurrence (see Table 1). This is supported by Ateweberhan et al. (2015) who emphasised that peak temperatures and high variability suppress growth and compromise the health of *K. alvarezii*, so mitigation requires cooler scheduling and careful site selection. The success of community-based projects depends on managing ecological constraints through appropriate site choice and planting windows to maintain optimal growing conditions.

At the same time, historical and contemporary literature frequently reports the presence of diverse bacterial assemblages in seaweed tissues exhibiting IID symptoms, particularly members of the family Vibrionaceae and other marine Gram-negative groups. Isolates such as *Vibrio*, *Pseudomonas*, *Aeromonas*, *Pasteurella*, *Edwardsiella*, *Plesiomonas*, and *Chromobacterium* have been recovered from bleached thalli, giving rise to early hypotheses that bacteria may act as causal agents transmitted through the water column (Aris and Labenua 2020; Astriana et al. 2024; Bessie et al. 2024; Tuhumury et al. 2024). However, research outcomes remain highly variable across locations and seasons. Many identifications are limited to the genus level using conventional culture-based approaches, and selective media that favour *Vibrio* spp. introduce methodological bias. These constraints complicate cross-study comparisons and limit consistent fulfilment of causality criteria required to substantiate a primary bacterial aetiology (Azis et al. 2022; Ward et al. 2021).

In this context, the accumulated evidence increasingly supports a pathobiome-based interpretation of IID, in which disease emerges from interactions between host physiology, microbial communities, and environmental stressors. Within this framework, abiotic factors, particularly thermal anomalies, reduced salinity, and nutrient limitation, play a central role in initiating physiological disturbance and microbial community shifts, while opportunistic microorganisms act as secondary contributors that exacerbate tissue damage rather than as sole aetiological agents (Ward et al. 2021). Field observations of mucus accumulation on diseased thalli are consistent with this interpretation and are attributed to the release of organic compounds, including carrageenan, following cell wall disintegration during bleaching events. These organic substrates subsequently provide a resource for opportunistic bacteria, accelerating symptom development without constituting evidence of a primary bacterial cause (Nasmia et al. 2014; Riyaz et al. 2021; Bessie et al. 2024).

Further support for an environmental trigger paradigm is provided by recent field studies demonstrating correlations between IID prevalence and symptom intensity with dissolved ammonium concentrations and rainfall, but not with other measured parameters such as temperature, salinity, pH, phosphate, nitrate, or nitrite (Tahiluddin and Terzi 2024). These findings should be interpreted cautiously, as increased rainfall is known to reduce salinity and enhance terrestrial runoff, thereby indirectly intensifying osmotic stress and altering dissolved organic matter availability. Taken together, the current corpus of evidence does not support attributing IID to specific bacterial taxa in the absence of environmental triggers. Instead, it favours a multifactorial, holobiont-level disease model in which

environmentally mediated host stress sets the stage for opportunistic microbial proliferation within a broader eco-physiological context.

From a practical perspective, this conceptual framing has direct implications for disease mitigation, positioning environmental stabilisation and context-sensitive risk management as primary priorities in seaweed aquaculture. In tropical regions, farmers commonly adjust cultivation calendars to avoid thermal extremes by suspending planting during periods of elevated sea surface temperatures and resuming production with the onset of the rainy season. Sanitation practices, such as pruning infected segments, can reduce inoculum pressure within farm units. However, indiscriminate disposal of infected fragments into surrounding waters may facilitate wider dissemination of pathogens, underscoring the need for enforced safe waste management procedures (Msuya 2020; Kambey et al. 2021b; Ward et al. 2021). Relocating farms to deeper waters has also been proposed as a means to stabilise temperature and reduce solar radiation exposure (Wenno et al. 2015; Pang and Liu 2019), yet this strategy is constrained by limited access, capital requirements, technical capacity, and the need to initiate new production cycles. Moreover, relocation does not uniformly guarantee reduced disease prevalence across ecological contexts. Indeed, Faisan et al. (2021) reported a trend toward higher IID prevalence in deeper waters, although the difference was not statistically significant, highlighting the importance of site-specific evaluation and continuous environmental monitoring (de Jong Cleyndert et al. 2021). Social dimensions within coastal communities, including gender roles, access to boats, and swimming ability, further influence the feasibility of such interventions and must be considered to avoid exacerbating participation inequalities (Ngajilo et al. 2025; Mwanyoka et al. 2025).

Overall, the convergence of ecological evidence and field experience reinforces the central role of abiotic drivers in shaping IID dynamics and indicates that effective mitigation should begin with environment informed risk management as the first line of defence. This should be complemented by high-resolution holobiont monitoring and multifactorial experiments that disentangle environmental triggers from bacterial cofactor roles. Such an integrated approach provides a more robust scientific basis for designing location- and season-specific interventions while enhancing the resilience of seaweed production systems under increasing climate variability.

Economic consequences and risks of diseases

Asian countries currently dominate global seaweed production, led by China, Indonesia, South Korea, and the Philippines. This dominance rests on favourable environmental conditions that include extensive coastlines, relatively low capital and resource requirements, and minimal dependence on groundwater or freshwater, rendering aquaculture highly relevant for coastal communities (Mariño et al. 2019; Mantri et al. 2024; Jaiswar et al. 2025). This relevance is reflected in the high involvement of coastal households; for example, more than 200,000 households in the Philippines and 267,000 in Indonesia rely on seaweed farming for their livelihoods (Langford et al. 2021; Faisan and Hurtado 2024). Despite these prospects, the sector faces serious biological pressures from disease outbreaks that directly suppress economic performance.

In tropical regions, IID is widely recognised as the most frequent disease affecting seaweed aquaculture, and it contributes to production declines. Its impact is evident in production trends in Africa. Total output dominated by eucheumatoid species, namely the genera *Kappaphycus* and *Eucheuma*, fell from 178,000 tonnes in 2015 to 81,000 tonnes in 2021

(UNCTAD 2024). In Tanzania, production in 2020 was approximately 11,000 tonnes with a value of about USD 2 million, down from 16,000 tonnes valued at around USD 4 million in 2015. One driver of this decline is IID, which can be exacerbated by contemporary climate change (Kunzmann et al. 2023). Epidemiologically, IID can reach very high prevalence, up to 99% in eucheumatoid farms (Largo et al. 2020), and even *E. denticulatum*, often regarded as more tolerant in earlier reports, has been affected (Pang et al. 2015; Ndawala et al. 2022). Unsurprisingly, the disease is acknowledged as a serious threat to the sustainability of farming in Tanzania (Msuya et al. 2022). Similar patterns appear across tropical Asia. In the Philippines, IID prevalence has been reported at 54% in Tawi-Tawi (Tahiluddin and Terzi 2024) and up to 100% in Mindanao (Faisan et al. 2021). Lower values for *Kappaphycus* in Eastern Samar likely reflect observational bias due to daily thallus cleaning by farmers and may not capture true infection intensity in the field (Yap-Dejeto et al. 2021). In Malaysia, particularly Sabah as a production hub, output declined by 1.7% from 182,000 tonnes in 2020 to 178,800 tonnes in 2021, and IID outbreaks have been indicated as one of the contributors to this reduction (DoF 2021; Kambey et al. 2021b). Financially, IID is identified as the principal disease in *Kappaphycus* and has been linked to local losses of approximately USD 17,300 to 18,500 per farm in Indonesia, as well as estimated annual losses of about USD 100 million in the Philippines (Cottier-Cook et al. 2016; Ward et al. 2020, 2021). Although national production in Indonesia and the Philippines remains high, disease pressure can erode profit margins, increase raw material supply volatility, and impede the attainment of growth targets for the sector. At policy and research levels, there is a need to harmonise surveillance protocols such as case definitions, severity scales, and loss reporting to conduct multi-season field trials across sites and to undertake microeconomic analyses at the farm unit level so that proposed interventions can be evaluated for both biological effectiveness and financial feasibility. Consequently, even though aggregate output in Indonesia and the Philippines remains substantial, trends in IID occurrence and its sensitivity to environmental change indicate that, without strengthened surveillance, standardised metrics, and adaptive farming practices, *Kappaphycus* production systems in tropical Asia are vulnerable to long-term productivity declines and supply fluctuations.

Losses arise not only in harvested quantity but also in carrageenan quality. Multiple studies show that carrageenan yield and gel strength in *K. alvarezii* and *E. denticulatum* are consistently higher in healthy biomass than in infected samples (Mendoza et al. 2002; Rantetondok and Latama 2017), consistent with findings that carrageenan quality declines in diseased seaweeds (Riyaz et al. 2021; Ndawala et al. 2022). Mechanistically, this quality degradation can be explained by two main processes. First, damage to the thallus surface, which is a key site for the synthesis and accumulation of cell wall polysaccharides, disrupts carrageenan biosynthesis (Ndawala et al. 2022). Second, enzymatic hydrolysis by pathogen-produced carrageenases cleaves carrageenan chains into lower molecular mass fragments, reducing gel strength and increasing losses during precipitation and extraction (Riyaz et al. 2022). As a consequence, infected biomass commands lower prices due to deteriorating hydrocolloid quality, which in turn can depress farm gate prices, shrink coastal household incomes, and heighten local economic instability. At the industry scale, chronic production disruptions can diminish market confidence and undermine the continuity of the global carrageenan supply chain. If outbreaks persist without effective intervention, some farming areas risk enterprise collapse (Khan et al. 2024). This context is particularly salient for major producers such as Indonesia, which has long supplied carrageenan raw materials to China as a global processing hub for the food, pharmaceutical, and biotechnological applications (Zhang et al. 2024a).

Disease pressure also compels farmers to adjust production strategies. A common response to increasing infection intensity is the temporary suspension of farming activities. This response should be understood not merely as a short-term operational adjustment, but rather as a consequence of structural constraints within coastal production systems, where few alternative commodities are simultaneously viable from both biophysical and economic perspectives. Ecologically, seaweeds require relatively narrow ranges of environmental conditions, particularly with respect to temperature and salinity, which substantially limit the spectrum of species that can be cultivated at a given site. For example, *S. japonica* is a cold-water seaweed, whereas *K. alvarezii* requires tropical environments. Switching species without altering environmental parameters is therefore often impractical or even impossible. These biological constraints are reinforced by market barriers. Well established market integration for carrageenan producing commodities keeps many farmers focused on *Eucheuma* spp., making diversification into other species risky in terms of market acceptance and profitability (Langford et al. 2022). Although *E. denticulatum* shows greater tolerance to IID and has a faster growth cycle of approximately 10 to 15 days compared with about 45 days in *K. alvarezii* (van Oort et al. 2025), typically fetches lower prices, around 10 to 50 PHP per kg in the Philippines and approximately IDR 6000 to 7500 per kg dry weight in East Nusa Tenggara, Indonesia, whereas *K. alvarezii* reaches 90 to 140 PHP per kg, about USD 0.69 to 1.09 per kg dry weight in India, and roughly IDR 20,000 to 22,000 per kg dry weight in Indonesia (Valderrama et al. 2015; Mantri et al. 2022; Tahiluddin et al. 2023; Tahiluddin and Roleda 2025b; Bessie et al. 2025). This price disparity makes diversification less attractive because profit margins are insufficient to sustain livelihoods, particularly for low-income coastal households. In several locations such as Zanzibar, where the majority of farmers are women, limited access to capital, market information, and trading networks further constrains adaptive options (Matoju et al. 2022). These lines of evidence indicate that disease in cultured seaweed species generates systemic economic impacts that not only erode local farm incomes but also threaten supply stability for downstream industries that rely on seaweed as a raw material.

Evidence from China and South Korea shows that both countries have suffered economic losses arising from disease outbreaks in seaweed cultivation. In *Pyropia*, disease outbreaks in China can cause regional yield losses of 25 to 30% (Gachon et al. 2010). Among these diseases, RRD is one of the most destructive, with substantial impacts on both yields and economic returns. For instance, yields reportedly declined by 84% in Fuding City, Fujian Province, China (Lai 2009). In January 2018, *P. yezoensis* declines were estimated at 50%, covering more than 7000 hectares in Haizhou Bay, Jiangsu Province, China (Yan et al. 2019). Since 2017, farmers in the Jiangsu production hub have also reported bleaching, with estimated losses of around USD 410 million in 2021, while pathogen identification was published only subsequently (BoF and NFTEC 2022; Liu et al. 2025b). Impacts extend to other commodities. National harvest volumes of *G. lemaneiformis* reportedly fell by at least 40% in 2023 compared with the previous year due to outbreaks of RD (Pang et al. 2024). Meanwhile, losses of 25 to 30% of regional harvest volumes of *S. japonica* in China have been attributed to diseases and pests as reported earlier (Wang et al. 2014), and these conditions are summarised in Table 1 and Fig. 1. Subsequently, *S. japonica* was affected by WSD with reductions in sporophyte length of up to 42% per day during 2017 to 2018, and the trend continued in 2021 when various *S. japonica* diseases were still reported along the Rongcheng coast of Shandong Province, the main production centre, causing large economic losses. The scale and persistence of these events were reported to be unprecedented in the past five decades (Wang et al. 2021; Zheng et al. 2025).

In South Korea, disease incidence is estimated to reduce *Pyropia* yields, including through RRD. Park et al. (2000) and Kim et al. (2014) documented declines in yield and prices of up to 15 to 20% and 20 to 30% in *Pyropia* farming areas in South Korea, respectively. Although affected seaweeds can still be harvested and marketed, lower quality and yields impose substantial negative economic impacts (Kim et al. 2014). In some farming seasons, GSD has caused substantial economic losses. For example, in 2012 to 2013, a GSD outbreak in the Seocheon sea farm in South Korea resulted in losses of about USD 1.1 million, comparable to losses from OD of USD 1.6 million (Kim et al. 2014). For *U. pinnatifida*, traditional practice generally spans October to April, but to avoid PD, farmers have shortened the farming period to December to February. This shorter harvest window reduces production per farm area. Park et al. (2008) showed that maximum growth in *U. pinnatifida* occurs between March and April and estimated that early harvesting to evade PD could reduce total biomass by roughly one third, equivalent to about 5 to 8 tonnes wet weight per hectare. The persistence of these diseases not only depresses annual yields but also undermines long-term productivity and the bioeconomic stability of aquaculture through supply uncertainty, rising mitigation costs, and risks of declining product quality. This evidence underscores that the economic burden of seaweed diseases is now apparent across all producing nations.

Key aspects of disease management in seaweed cultivation

Biosecurity measures

The presence of diverse pathogens whose pressure is increasing due to climate change, together with abiotic factors such as water movement and light intensity, has the potential to amplify disease risk in seaweed aquaculture (Ward et al. 2021; Zhang et al. 2024b). This situation underscores the urgency of strengthening biosecurity protocols as the first line of defence. Internationally, biosecurity is recognised as a key approach to minimise the introduction and spread of disease within the aquaculture industry. To be effective, implementation requires interdisciplinary governance and regulatory mechanisms at national and international levels (Campbell et al. 2020), particularly in the context of increasingly intensive cross-border trade (Bray et al. 2024). At the national level, strategies must be tailored to local challenges while being aligned with the development of progressive management pathways for the seaweed sector (Cottier-Cook et al. 2022). Clear, simple, and integrated strategies are expected to raise vigilance, promote the adoption of sustainable management systems, and streamline research and policy development by reducing cross-sector redundancies (Hulme 2021). This urgency is sharpened by the growing interconnectedness between aquaculture production and global trade flows.

In practice, current biosecurity measures focus on prevention, treatment, and mitigation of disease, while truly effective control remains a central challenge (Bray et al. 2024). This is understandable because seaweed is cultivated in open waters that are difficult to control, unlike terrestrial crops or aquatic animals in ponds that grow in more controlled environments. Nevertheless, field evidence shows that simple interventions can materially reduce risk. A case study in Sabah, Malaysia, showed that the use of healthy propagules and the cleaning of thalli and ropes can reduce disease spread in seaweed farms (Kambey et al. 2021b). Evaluation of biosecurity implementation should be conducted through systematic analysis of policy frameworks, regulations, and national standards. In Indonesia, identified

barriers include a limited range of biosecurity approaches, low uptake of modern technologies, and inadequate prevention and treatment procedures (Kambey et al. 2020). A relevant example comes from farming experiments in South Sulawesi, a major seaweed production hub in Indonesia (Alam et al. 2025), which documented the reuse of biomass previously exposed to IID after only separating it from infected thalli, washing it, and then replanting it in the next cycle (van Oort et al. 2025). This practice increases the likelihood of disease persistence and transmission into subsequent production cycles and reflects weak ecology and technology-based mitigation at the farm level. Conversely, correct application of biosecurity principles including the removal of infected seaweeds in the field has been shown to improve seaweed health and reduce disease impacts by up to 83% in treated plots (Sondak et al. 2025). In practice, removal of thalli is often conducted by cutting affected segments and allowing apparently unaffected portions to regenerate and regrow, although overall productivity declines (Hurtado et al. 2019).

These findings provide an evidence-based foundation that biosecurity plays a central role in preventing and mitigating seaweed diseases while supporting improvements in both quantity and quality of production. Accordingly, investment in farm-level biosecurity practices can generate economic gains through reduced outbreak losses and increased yields. In addition, more detailed and coherent biosecurity policies are needed to support the growth of the seaweed industry and deliver global benefits through cross-border policy integration. Knowledge, attitudes, and practices surveys, or KAP, can be used to design effective policies and management strategies to address biosecurity risks in the seaweed aquaculture sector. This method, which has been applied in Malaysia, is expected to improve industry governance and support sustained productivity (Kambey et al. 2021a).

Although aquaculture has long been practiced, the sector is also a major pathway for the spread of alien species in aquatic ecosystems. To date, there is no dedicated health code for aquatic plants nor a primary international mechanism designed to control diseases associated with the trade of seaweed-based organisms and aquaculture products. In other words, there are no globally recognised cross-national standards for reporting, diagnosis, or control of seaweed disease outbreaks. This gap causes many outbreaks to escape global attention, to be managed locally, and often to remain poorly documented (Cottier-Cook et al. 2021). Concerns intensify when lesser-known pathogens are inadvertently introduced through agricultural and horticultural trade or through the use of seedstock sourced from outside the country. In 2005, in an effort to rescue the seaweed farming industry in Kudat, Sabah, Malaysia, which was severely affected by IID, seedlings were imported from southern Philippines. Rather than solving the problem, this action exacerbated the situation by introducing pathogens that triggered more severe outbreaks such as secondary infections and thallus breakage due to inadequate monitoring and quarantine (Vairappan et al. 2008). Similar concerns were raised by Badis et al. (2020b), who concluded that international movement of seaweed seed without monitoring or biosecurity regulation could form the basis of a serious ecological and economic crisis in marine environments.

This contrasts with the aquatic animal sector, which is regulated by the Aquatic animal health code, developed by the World Organisation for Animal Health (WOAH, formerly the Office International des Epizooties). This code is part of the legal framework of the World Trade Organization (WTO) and governs measures for the prevention, early detection, reporting, notification, and control of pathogens in aquatic animals, aimed at preventing their spread through international trade. The consequences of regulatory gaps in the seaweed sector are illustrated by documented cases from Indonesia, Tanzania, and the Philippines, as well as reports of unauthorised cultivation of *S. japonica* in southern Chile, a species that is non-native to the region. In this case, farming activities reportedly

occurred outside established approval processes for the introduction and cultivation of alien seaweeds, highlighting weaknesses in biosecurity governance related to species translocation and bioinvasion risk (Camus et al. 2022). These lessons are particularly pertinent for jurisdictions assessing the development of aquaculture based on new or expanded uses of seaweed species. For example, while *Hormosira banksii* is a common intertidal seaweed naturally distributed throughout temperate New Zealand waters, any proposal to scale up its utilisation for aquaculture purposes still necessitates robust biosecurity frameworks (Loy and Jeffs 2025). Priorities include precautionary frameworks through pre border risk assessment, quarantine procedures and certification of seed sources, and ecological and genetic monitoring. Accordingly, clearer and more comprehensive regulations at national and international levels are urgently needed. Collaboration through a One Biosecurity approach has the potential to integrate fragmented frameworks and to better account for cross sector risks and stakeholder needs (Bray et al. 2024).

To address these gaps, FAO, in collaboration with aquaculture experts, developed the Progressive Management Pathway for Improving Aquaculture Biosecurity for Seaweed (PMP/AB-Seaweed), a framework specifically designed to address biosecurity challenges in seaweed aquaculture (Cottier-Cook et al. 2022). This pathway emphasises the formation of cross-sector task forces spanning public and private stakeholders, the development of a National Strategy for Seaweed Health Management, the establishment of standardised diagnostic, reporting, and response protocols, and the integration of seaweed health governance into national fisheries or agricultural policies. Collectively, these components aim to operationalise a One Health approach. Although by 2025 seaweed-specific international standards remain under finalisation, this initiative represents an important step toward harmonised disease management at the global level. Strengthening the scientific basis remains urgent, including the clear establishment of disease aetiology and the further development of diagnostic and therapeutic techniques that are still at an early stage, so that policy is not formulated without adequate evidential support (Cottier-Cook et al. 2022). Insights from the holobiont perspective indicate that managing beneficial microbiota can enhance resilience (Kambey et al. 2021b). Looking ahead, probiotics, breeding, and genetic approaches for disease-resistant varieties have the potential to complement conventional biosecurity.

The use of inorganic materials

The application of inorganic agents in seaweed aquaculture has strengthened in recent years. This trend has been largely driven by the need to enhance productivity through accelerated biomass accumulation and shortened harvest cycles (Roleda et al. 2025). In line with this objective, nutrient enrichment using inorganic fertilisers such as NPK has been employed to stimulate growth, improve tolerance to nutrient stress, and suppress the incidence of IID in eucheumatoid seaweeds (Luhan et al. 2015; Tahiluddin and Terzi 2024; Muyong and Tahiluddin 2024). Field evidence has strengthened the case for this approach. Muyong and Tahiluddin (2024) demonstrated that ammonium and phosphate supplementation in *K. alvarezii* provided complete protection against IID across multiple cultivation methods tested in Tawi-Tawi, Southern Philippines. The practice was subsequently extended to other species, particularly *K. striatus*, which has been empirically reported to be more susceptible to IID than *K. alvarezii* and *E. denticulatum* (Tahiluddin and Eldani-Tahiluddin 2024). Collectively, these findings indicate that inorganic interventions can serve as a critical component of seaweed health management when applied appropriately and under controlled conditions.

However, these technical benefits are accompanied by ecological, economic, and product-quality considerations that require careful attention. From an environmental perspective, the use of synthetic inputs may stimulate eutrophication, alter the balance of coastal microbiomes, and increase pollutant loads in and around aquaculture waters (Liu et al. 2020; Lähteenmäki-Uutela et al. 2021; Mancuso et al. 2023). From a production standpoint, although inorganic fertilisers can accelerate eucheumatoid growth, they frequently encourage premature harvesting before tissues reach full maturity. This can reduce gel strength and carrageenan quality, ultimately lowering yield and product value (Roleda et al. 2025; Abduraup et al. 2025).

The economic implications are equally significant. Dependence on commercial inorganic fertilisers is incompatible with international organic certification standards, thereby limiting access to premium markets and potentially weakening the competitiveness of high-value seaweed products (Gómez and Lähteenmäki-Uutela 2021). Across the value chain, these effects extend from primary production to downstream marketing. The use of inorganic nutrient enrichment can disqualify seaweed biomass and its derivatives, including carrageenan, from organic certification—particularly within food and feed production chains (Tahiluddin and Roleda 2025a). Consequently, upstream production decisions directly shape downstream market positioning.

These implications are particularly relevant for major eucheumatoid-producing countries such as the Philippines, especially the provinces of Tawi-Tawi and Sulu, where inorganic nutrient supplementation remains in cultivation (Tahiluddin et al. 2023). These regions must reassess the balance between productivity-driven strategies and compliance with target market standards. Recent literature advocates a transition toward more sustainable approaches through the integration of strict biosecurity, improved seedstock quality, and the use of biofertilisers aligned with organic guidelines as alternatives to synthetic inputs (Kambey et al. 2021b; Liu et al. 2024b; Alam et al. 2025). Achieving this transition will require coordinated policy support, sustained research investment, and farmer extension programmes. The overarching goal is to develop farming systems that remain productive, meet certification requirements, safeguard the integrity of coastal ecosystems, and maintain access to high-value markets. Such an integrated approach enables effective disease control without compromising organic legitimacy or the economic value of seaweed products.

Genetic diversification

Genetic diversification is the foundation of population genetics approaches to enhance productivity, quality, and disease resistance in seaweed aquaculture. In many major seaweed commodities, particularly *Kappaphycus* and *Eucheuma*, the predominance of vegetative propagation, limited domestication, and the reuse of harvested biomass as seedstock have produced farmed populations with low genetic diversity, a condition associated with reduced growth rates, suboptimal morphology such as thinning thallus branching, and heightened susceptibility to pathogens, as reported by Sundarraj et al. (2025) and Gonzaga et al. (2025) on eucheumenoid populations in tropical regions. From a population genetics perspective, prolonged cloning reduces effective population size, which in turn diminishes genetic diversity or allelic richness and narrows the portfolio of genetic material available to future generations and breeding efforts (Lim et al. 2017; Thomson 2021).

Efforts to broaden the genetic base through selection of local varieties and interstrain crosses are logical first steps, yet the complexity of disease pressure and environmental variability demands more targeted strategies grounded in population data. Monitoring parameters such as heterozygosity, allelic richness, and among population differentiation provides a diversity map that can be translated into seed decisions (Van Rossum and Hardy 2022), for example complementary crossing pairs, multi-site seed sources, and mosaic planting across genotypes to disrupt pathogen transmission. In Malaysia, the discovery of new haplotypes in wild *Kappaphycus* populations reveals untapped reservoirs of adaptive alleles that can be introduced into farmed populations to strengthen long-term resilience (Tan et al. 2022). This approach underscores the integration of population genetic surveys, germplasm management, and breeding programmes so that adaptive alleles flow into culture stocks in a measured way.

Compelling applications come from heteromorphic life cycle species at temperate latitudes. Hybridisation programmes for *S. japonica* and *U. pinnatifida* in China and South Korea leverage the gametophyte phase to produce F1 cultivars through controlled crosses. Goecke et al. (2020) reported *Saccharina* hybrids with greater thallus size and wet weight than parental lines, indicating heterosis relevant for productivity. Intraspecific and interspecific hybridisation can also combine adaptive alleles that enhance tolerance to thermal stress, a key trait under ocean warming. Downstream, farmer concerns in Dongtou District, China, about declining performance of older varieties that have become increasingly disease prone highlight the urgency of genetic renewal of culture stocks through planned crossing schemes (Liu et al. 2025a). In line with this, South Korea has developed *P. yezoensis* Gold 1, which shows high resilience to RRD and has been widely commercialised, underscoring the value of locally based domestication and breeding to sustain high-value commodities (Hyun et al. 2023).

Advances in biotechnology expand the scope of intervention without departing from population genetic principles. Marker-assisted breeding and, where available, genomic selection can accelerate the identification and introgression of resistance alleles into superior genetic backgrounds. Recent evidence shows selectively developed lines that express stronger immune responses to pathogen infection than conventional varieties, reinforcing the value of data-driven targeted breeding (Zhuang et al. 2024). At the same time, specific crosses that exhibit heterosis not only increase growth rate but also broaden tolerance thresholds to temperature fluctuations, which indirectly reduces vulnerability to opportunistic diseases triggered by environmental stress (Song et al. 2024). Integrating information on genetic diversity, crossing design, and phenotypic evaluation of resistance therefore yields breeding pipelines that are more predictive and efficient.

Exploration of wild germplasm remains a pillar for adding variation, especially when local cultivars show narrowing diversity. Findings from Malaysia confirm limited diversity in local cultivars and the strategic value of prospecting wild populations as sources of new adaptive alleles for *Kappaphycus* spp. and *E. denticulatum* that confer tolerance to high temperature and pathogen infection (Tan et al. 2022; Yahya et al. 2024). Even so, recent literature cautions that translocation of genetic material across locations or species must be managed within strict biosecurity frameworks. Risks of genetic pollution and the emergence of novel pathogen strains should be anticipated through genetic screening, multi-site trials, and ecological impact assessment prior to varietal release, as highlighted by recent ecological and policy reviews (Spillias et al. 2022; Crisostomo and Roleda 2024; Roleda et al. 2024; Ngwaerobo et al. 2025).

Overall, population genetics provides an operational set of concepts and metrics to guide improvements in disease resistance in cultivated seaweeds. Mapping population

structure and connectivity, quantifying genetic variation with molecular markers, and managing gene flow between wild and farmed stocks enable the design of seed systems with substantial genetic diversity. When combined with mosaic planting across genotypes, rotation of seed sources, and cross-environment resistance evaluation, this approach not only enhances agronomic performance but also slows pathogen adaptation. Looking ahead, research should integrate regional population genetic surveys, resistance marker panels, and standardised trials so that domestication yields cultivars that are robust and competitive. In line with this trajectory, Gacura et al. (2026) reported new *K. alvarezii* haplotypes that grow rapidly and are less susceptible to IID, which are now under multi-location testing including in Bohol, Philippines.

Effectiveness of biostimulants

Biostimulants are defined as active compounds that are non-nutritional and non-pesticide which modulate physiological processes in plants and offer substantial potential to enhance growth while strengthening tolerance to environmental stress. In recent years, their application in seaweed aquaculture has attracted global research attention as a sustainable strategy to increase productivity and resilience to disease. One widely studied biostimulant is the extract derived from the brown seaweed *Ascophyllum nodosum*, which has shown notable efficacy in supporting large scale seedling production to address bottlenecks in seedstock supply. Compared with liquid extracts such as those from *A. nodosum* (Borlongan et al. 2024), *Ascophyllum* marine plant extract powder or AMPEP is more widely used and has demonstrated superior performance in optimising seaweed growth even under suboptimal environmental conditions (Hurtado et al. 2024). For example, application of AMPEP to *S. latissima* significantly accelerates growth rates at elevated temperatures that typically limit production (Hurtado and Critchley 2020), underscoring its potential to mitigate climate related constraints in seaweed farming. However, biological responses to AMPEP are species specific and conditioned by environmental parameters. A study by Shin et al. (2024) showed that although AMPEP did not increase thermal tolerance in *P. yezoensis*, it triggered reproduction through the release of archeospores at low temperatures. These findings encourage exploration of alternative biostimulants such as *Sargassum horneri* extracts that may support thermal resilience in particular species (Han et al. 2023). In addition, Zainuddin (et al. 2019) reported that dipping seedstock of seaweeds such as *G. verrucosa* in *Caulerpa* extracts prior to pond culture is a promising step that warrants further testing to reduce damage from IID in seaweed aquaculture. This aspect merits closer examination in relation to infection dynamics since thermal stress at high temperature can heighten susceptibility to opportunistic pathogens as also noted by Zainuddin et al. (2019).

Beyond their role in promoting growth and tolerance to environmental pressures, AMPEP has also proven effective as a prophylactic agent against disease. Significant reductions in IID incidence have been observed in *K. alvarezii* treated with AMPEP solution during the summer period (Munisamy et al. 2023; Capacio et al. 2024). This effectiveness is consistent with evidence that AMPEP can activate natural defence pathways in the red seaweed *K. alvarezii* against pathogens and environmental change (Loureiro et al. 2012). From an economic standpoint, AMPEP is affordable for smallholder farmers in developing countries. In India, for example, an additional production cost of about USD 0.066 per kg of dry *Kappaphycus* in 2021 was associated with a yield increase of 16.66% following AMPEP application (Shanmugam and Seth 2018;

Munisamy et al. 2023). Nevertheless, inconsistency among batches, likely due to variation in chemical composition or extraction methods, can reduce effectiveness and therefore calls for standardisation of production processes (Hurtado et al. 2024).

Alongside seaweed extract-based biostimulants such as AMPEP and liquid *A. nodosum* products, functional microbe-based approaches also show considerable promise for enhancing disease resistance. In China, probiotic *P. piscicida* strains (P3 and P6), especially when combined with short-term drying, effectively suppressed the development of *P. porphyrae*, the pathogen that causes RRD in *Pyropia*, without damaging thalli (Weng et al. 2024). A holobiont-oriented approach that leverages symbiotic interactions between seaweeds and associated microbes further reinforces this concept. The use of *V. alginolyticus* strain X 2 reduced bleaching in *S. japonica* and supports the development of seaweed beneficial microorganism frameworks for cultivation (Ma et al. 2023). These findings indicate the potential role of combining probiotics and biostimulants in supporting more resilient and environmentally sustainable farming practices.

Although biostimulants hold substantial promise, their effectiveness is shaped by environmental factors such as temperature, salinity, and season. Optimal implementation therefore requires tuning of applications to local conditions. Large-scale deployment also faces challenges that include product inconsistency, limited long-term datasets, and potential ecological effects such as shifts in marine microbiomes, which require systematic evaluation through multi-site field trials and extended monitoring (Brakel et al. 2021). As environmentally friendly alternatives to synthetic inputs, biostimulants align with modern biosecurity principles and contribute to the sustainable transformation of global aquaculture (Jaiswar et al. 2025).

Application of co-culture

Co-culture of seaweeds with other aquatic organisms has emerged as an innovative approach in sustainable aquaculture with the aim of establishing symbiotic relationships that support ecosystem stability and multi-species productivity. This approach mimics natural processes in coastal systems in which nutrient wastes generated by animals are utilised by seaweeds, while the presence of seaweeds helps stabilise water quality through nutrient uptake and oxygen production (Zhang et al. 2025). Although its ecological potential is recognised, much of the earlier literature has focused on positive effects for companion organisms such as shrimp, crabs, fish, and sea cucumbers, whereas impacts on the physiology and health of farmed seaweeds remain relatively overlooked (Pintado et al. 2023; Anh et al. 2024). For example, integrating *Penaeus indicus* with *Gracilaria tenuistipitata* increased shrimp growth rate and immune responses (Das et al. 2022), and co-culture of *Ulva pertusa* and *Gracilariopsis* with scallops improved water quality, suppressed proliferation of *Vibrio* spp., and strengthened bivalve immunity, although evidence on direct implications for seaweed health is still limited (Zhang et al. 2025). Recent work shows that co-culture of *Gracilariopsis* with *Penaeus monodon* did not alter agar content or gel strength and even increased the specific growth rate of seaweeds, yet health parameters should be evaluated explicitly so that growth health trade-offs can be assessed accurately and productivity gains remain aligned with sustainable farming (Nauta et al. 2025).

Co-culture provides a range of ecosystem services relevant to disease control, including pathogen filtration, nutrient regulation, oxygenation, and microbial competition, which collectively enhance biosecurity and system resilience (Zhang et al. 2025; Zaat et al. 2025). In an intraspecific context, coexistence of *S. horneri* with *G. lemaneiformis* stimulates

seaweed growth while maintaining a balanced biomass ratio that supports community stability (Song et al. 2022). Co-culture among seaweeds such as *K. alvarezii*, *K. striatum*, and *E. denticulatum* has also been reported to reduce the incidence of IID compared with homogeneous monoculture systems (Pang et al. 2015; Bessie et al. 2024). In pond-based systems, combining *G. verrucosa* and *C. racemosa* shows potential to lessen damage from IID and warrants further testing at operational scale to confirm consistency of benefits across sites (Zainuddin et al. 2019). Taken together, these findings indicate that species diversification in co-culture can create a mini ecosystem that is more robust to external disturbance.

Even so, co-culture is not risk free for seaweed health. Higher prevalence of IID in *Kappaphycus* within integrated multi-trophic aquaculture alongside sea cucumbers and oysters indicates that combined thermal stress and nutrient deficiency can trigger host vulnerability when environmental management is inadequate (Bernard et al. 2024). Physico-chemical conditions such as temperature and nutrient availability are therefore critical to co-culture success. The presence of non-target species can also act as pathogen reservoirs. *Bangia fuscopurpurea*, which can coexist with farmed seaweeds, has been identified as a potential reservoir of oomycetes that cause disease in *P. yezoensis*, implying that non-target species may worsen disease dynamics even if they are not a direct component of the co-culture system (Park et al. 2025). Furthermore, co-culture of *G. lemaneiformis* with *S. japonica* has been reported to increase the risk of pathogen spread unless supported by strict risk management at operational and biosecurity levels (Pang et al. 2025).

Several mitigation strategies have been proposed to minimise these risks. In Europe, probiotic colonisation of *Ulva* by *Phaeobacter* in co-culture with fish produced anti pathogenic compounds that significantly reduced disease incidence, with effectiveness contingent on microenvironment management including light cycle regimes to stabilise microbial communities (Del Olmo et al. 2025). In parallel, selecting physiologically compatible species is essential to suppress infection risk, so community design should consider nutritional requirements, temperature tolerance, and host microbiome dynamics (Zaat et al. 2025). In the context of climate adaptation, *Ulva compressa* can facilitate the growth of *Ulva lacinulata*, which is sensitive to temperature fluctuation, indicating that intraspecific co-culture can be a viable strategy to enhance thermal resilience in seaweeds (Bermejo et al. 2023).

Overall, the available evidence indicates that co-culture can become a future farming model that is productive, environmentally friendly, and resilient to climate variability and disease pressure. Successful implementation requires science-based design, careful species selection, and stringent pathogen control and biosecurity protocols. Continued research is needed to refine this approach and to ensure that ecological gains and productivity improvements are achieved without compromising the health of farmed seaweeds. Once the scientific and operational foundations are established, the next priority is to promote commercial scale adoption of polyculture, noting that farmers, particularly in developing countries such as Indonesia, require support to adopt co-culture practices, including strengthened bargaining positions with intermediaries and access to finance that enables initial investment (Azhari et al. 2026).

Multisectoral collaboration

Diseases affecting farmed seaweed commodities constitute a global challenge that demands a coordinated, cross-sectoral response. A compilation of disease reports in cultivated seaweeds up to 2025 (Table 1) indicates diverse outbreaks across major producer nations. The

magnitude of the economic losses and social impacts, particularly for millions of coastal farmers, underscores that disease management cannot rest with any single actor. Multisector collaboration involving government, the scientific community, industry, local communities, and international organisations is required to strengthen prevention, early detection, reporting, and control throughout the seaweed supply chain (Jompa et al. 2024). From a scientific standpoint, strengthening the foundations of seaweed pathology is central to collaboration. Genomic and metagenomic surveillance at farm sites, integrated with systematic recording of symptom phenotypes, severity, and transmissibility, accelerates pathogen identification; meetings and workshops help standardise protocols, share resources, and align analytics, thereby improving diagnostics, guiding evidence-based control strategies, and deepening understanding of interactions between hosts and pathogens. From the perspective of policy and governance, interdisciplinary approaches are needed to design management strategies, farming practices, and regulatory frameworks that minimise disease risk while supporting the sustainable development goals, especially SDGs 14 on life below water (Spillias et al. 2022; Vadassery and Neish 2024). Empirically grounded social-ecological frameworks offer a lens for assessing trade-offs and co-benefits in decision making across actors (Cerca et al. 2023). At the implementation level, participatory methods such as group model building facilitate stakeholder engagement, surface critical sustainability issues, and map leverage points for intervention, including those related to disease impacts in seaweed aquaculture systems (Muflikh et al. 2024).

Operationally, multisector collaboration should be translated into clear workflow mechanisms. First, two-way reporting channels at the farmer level should be strengthened for any anomalies indicating emerging disease. Second, rapid scientific responses are needed through aetiological investigation, risk assessment, and evidence-based recommendations. Third, relevant authorities should formulate adaptive policies for prevention, control, and impact mitigation, including guidance on good aquaculture practices, biosecurity, and post-harvest management. Consistent feedback loops between the field, laboratory, and regulators ensure continuous learning and progressively more precise responses over time. Data infrastructure is a critical node connecting all these elements. Standardised documentation of pathogen diversity, biogeography, host range, outbreak dynamics, and cross-sector metadata should be developed into a globally accessible database via a web portal to expedite detection, enhance cross-study comparability, and strengthen response planning (Strittmatter et al. 2022). Applying principles of data openness and interoperability will bolster transparency, replication, and knowledge transfer across regions (Mwanyoka et al. 2025). To accelerate collaboration and capacity, institutional platforms play a strategic role. For example, the Government of Indonesia's initiative to establish the international tropical seaweed research centre (ITSRC) could serve as a hub for research, technology transfer, and capacity development. Through a coordinated multi-stakeholder network, the ITSRC has the potential to foster a resilient research ecosystem capable of addressing disease challenges while promoting sustainable growth of the seaweed industry. Taken together, integrating scientific, policy, operational, data, and institutional dimensions within a single multisector collaborative framework offers a realistic and effective pathway to reduce the disease burden in seaweed aquaculture.

Application of pathogen and disease detection methods

By 2025, the integration of cutting-edge technologies, rigorous research methodologies, and cross-national collaboration produced increasingly sensitive and accurate tools to

identify pathogens in farmed seaweeds. Applying these methods has enhanced biosecurity and production stability through early detection, which is crucial for preventing economic losses. Despite this progress, knowledge gaps concerning viruses, particularly their ecology and diversity, remain evident. Rather than a reason to delay intervention, these limitations underscore the urgency of further research to strengthen detection and mitigation strategies. Pioneering studies have begun to address this gap. Small interfering RNA-based approaches have been used to detect novel RNA and DNA viruses (Dekker et al. 2024), while virome metagenomic sequencing has been applied to characterise viral communities, including those associated with BD in *Ulva* (van der Loos et al. 2023). Given that this topic was relatively underexplored prior to 2020, these findings represent important milestones, although causal links between viruses and clinical symptoms still require experimental validation.

In Asia, research and field monitoring have focused more on bacterial and parasitic pathogens. Histology, with and without staining, has been employed to assess pathogen impacts at the tissue level (Riyaz et al. 2021; Wang et al. 2021). However, reliance on morphological change makes early detection challenging, since lesions often become apparent only at advanced stages. Accordingly, integrating higher resolution microscopy such as SEM and TEM with molecular analyses and other omics such as metabolomics has been proposed to deepen understanding of seaweed pathology and accelerate diagnosis (Kwak et al. 2017; Ward et al. 2020; Kim et al. 2014). At the molecular level, several breakthroughs have improved diagnostic accuracy and speed. Quantitative PCR or qPCR enables high precision quantification of pathogens (Ferrari et al. 2021; Liu et al. 2024a) and PCR restriction fragment length polymorphism or PCR RFLP based on the mitochondrial cytochrome c oxidase subunit 2 cox2 has been reported as effective for detecting RRD in *Pyropia* (Lee and Lee 2022). Previously, highly specific cox1 and cox2 markers for *Pythium* were developed for long-term monitoring of infection and distribution patterns of individual *Pythium* in *Pyropia* aquaculture ponds, thereby providing information to predict infection and prevent mass mortality due to RRD (Lee et al. 2017; Qiu et al. 2019). In addition, nanotechnology likewise offers opportunities for detection at very low concentrations using devices that are cost efficient and easy to operate (Khan et al. 2023). Gold nanoparticle based nanobiosensors provide rapid tests through changes in plasmonic properties that indicate the presence of viral particles. This approach has been successfully applied to viruses of land crops and is expected to be adaptable to seaweed viruses (Kulabhusan et al. 2022; Gurung et al. 2023).

Artificial intelligence is expanding the horizon for real-time monitoring of seaweed health. Computer-vision systems based on machine-learning approaches, including a robotic platform employing the k-nearest neighbours algorithm in Indonesia, have demonstrated the ability to identify IID from thallus colour and texture features with a field accuracy of 86.7% (Saputro et al. 2024). This approach enhances detection efficiency while reducing reliance on manual inspection, making it particularly relevant for medium to large-scale farming operations. Integration with the Internet of Things (IoT), through underwater cameras, unmanned vehicles, and environmental sensors, further enables the continuous collection of temperature, salinity, nutrient, and biomass data. Such systems support early detection of physiological stress or infection and facilitate data-driven decision-making. In parallel, optical remote-sensing technologies are emerging as key components of early disease detection. Hyper- and multispectral sensors allow the measurement of thallus reflectance across multiple wavelengths, enabling infection-related colour changes to be detected before they become visually apparent.

For example, Alevizos et al. (2024) utilised drone-based reflectance data and vegetation indices combined with machine-learning algorithms to distinguish healthy and IID-infected *K. alvarezii* and *E. denticulatum*, achieving classification accuracies ranging from 83 to 99% based on spectral signatures and selected indices. Although the spatial resolution of satellite imagery remains insufficient for mapping individual seaweed farms (Kellaris et al. 2019), remote sensing nonetheless holds promise as an early-warning system for large-scale or extraordinary events, such as regional mass bleaching episodes. Moreover, low-cost underwater stereo-camera systems integrated with computer-vision techniques have successfully been applied to monitor *Saccharina* growth in Europe, and similar approaches could be adapted to automate the detection of disease symptoms, including necrotic lesions or epiphyte overgrowth, using machine-learning frameworks (Gerlo et al. 2023).

Looking ahead, mainstreaming these technologies into integrated disease management should be a priority agenda. Strategies need to combine preventive, curative, and predictive approaches tailored to local farming environments. Successful implementation depends on standardisation of terminology and protocols, cost reductions to ensure affordability for smallholders, and improved usability in remote settings with limited infrastructure (Cottier-Cook et al. 2021; Strittmatter et al. 2022; Murúa et al. 2023). Overall, by 2025, there is a clear shift toward proactive, sensor and data-driven seaweed health management. With continued refinement spanning molecular methods such as PCR, qPCR, and LAMP, nano-biosensors, artificial intelligence connected to the IoT, and optical imaging, the seaweed industry is poised to reduce outbreak risks, increase yields, and safeguard the livelihoods of millions of coastal farmers (Cottier-Cook et al. 2021; Alevizos et al. 2024). Challenges in standardisation, cost, and device robustness remain to be addressed, yet the current trajectory suggests the sector is ready for broad adoption.

Conclusion and future outlook

As of 2025, global seaweed production has experienced disease outbreaks of increasing frequency and severity, with at least 16 distinct diseases identified (see Table 1). In tropical waters, IID remains the most frequently reported disease affecting cultivated seaweeds. In contrast, in temperate to cold-water regions, the disease profile is more diverse, with multiple forms of rot and spot diseases representing a major source of diseases and posing challenges to the productivity of seaweed aquaculture in these geographical areas (Fig. 1). Outbreaks tend to spread rapidly and are often triggered by environmental stressors (see Fig. 3). The open-sea farming context limits control over environmental variables. Even so, the past years have brought substantial advances in understanding aetiology, transmission dynamics, and mitigation options. Persistent barriers arise from gaps in official reporting and the absence of international standards for outbreak identification and response. At the operational level, farmers often rely on unvalidated empirical actions, including the disposal of infected seaweeds or the use of inorganic inputs. Such measures may yield short-term benefits yet risk long-term ecological impacts.

Amid accelerating climate change, the need to strengthen biosecurity is increasingly urgent. Reports of losses that can reach 100% of production when outbreaks go unchecked underscore the scale of risk to farmers and supply chains (Suyo-Diala and Hurtado 2024). Evidence of pathogen invasion in wild red seaweeds in Chile highlights the importance of systemic surveillance and risk-based interventions across the aquaculture value chain, from seedstock to product distribution (Murúa et al. 2024). Multi-stakeholder collaboration is

essential, an exemplary case of such collaboration is the Centre of Excellence for Seaweed (PUIRL, formerly Centre of Excellence for Development and Utilisation of Seaweeds) at Hasanuddin University, Indonesia, which brought together academics, government entities, and farmers in the implementation of innovative seaweed farming practices (Kasmiati et al. 2024). The complexity of disease biology, which often exceeds expectations, demands continuous innovation and research designs that respond to day-to-day farming realities, including local environmental variability and the dynamic behaviour of pathogens (Faisan et al. 2021).

Molecular markers are beginning to support the selection of disease-resistant lines without eroding germplasm diversity, enabling breeding programmes to proceed in a more targeted way while safeguarding the diversity needed for long-term resilience (Tan et al. 2024). Moreover, modern technologies in monitoring seaweed health, such as remote sensing, are becoming increasingly crucial for early detection of potential disease outbreaks. Proximal optical sensing enables the identification of symptoms with high accuracy, allowing interventions before declines in farm performance become visible (Alevizos et al. 2024). With ongoing development and evaluation, these innovations are anticipated to play a pivotal role in addressing disease challenges and promoting the establishment of sustainable aquaculture in the future.

The future research and policy agenda should align several complementary priorities within a coherent workstream. Standardisation of terminology, diagnostic criteria, and reporting protocols needs to be accelerated so that data across locations can be compared and acted upon without delay. Risk-based biosecurity must be reinforced from seedstock through post-harvest via quarantine, health certification, and sustained environmental monitoring. Diagnostic technologies that are accessible to small-scale farmers should be scaled, including portable molecular platforms, optical sensing, and early warning systems that use environmental data. Breeding and selection programmes should deliver lines with stable resistance across environmental conditions while maintaining genetic diversity. Collaborative governance that bridges laboratories, farmers, government, and international networks should be strengthened through participatory mechanisms such as My Seaweed Looks Weird (MSLW) web portal to accelerate community reporting and response, together with enhanced local diagnostic capacity and transparent cross-border reporting to speed collective learning and reduce transboundary risk (Strittmatter et al. 2022).

With a combination of intensive research, context-appropriate technological innovation, and consistent policy support, the seaweed industry can grow in a sustainable, productive, and resilient manner. Illustrative implementation that balances economic objectives, social well-being, and environmental integrity is emerging in *K. alvarezii* farming in India, which demonstrates an integrated approach to productivity gains and system resilience (Mantri et al. 2024). Ultimately, success in controlling seaweed diseases is determined not only by the sophistication of technology but also by the quality of collaboration, the consistency of policy, and the commitment of an innovation ecosystem oriented toward sustainability.

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Authors and Affiliations

Ade Wahyul Fajri Alimin[1,9] · Kasmiati Kasmiati[2,9] · Aswin Aswin[3] · Fatchurrachman Fatchurrachman[4] · Nur Yanti Pangloli[5] · Dea Pramita[2] · Winda Wijaya[6] · Rahman Romli[7] · Elmi Nurhaidah Zainuddin[2,9,10] · Jamaluddin Fitrah Alam[8,9]

✉ Ade Wahyul Fajri Alimin
adalimin31@gmail.com

✉ Kasmiati Kasmiati
kasmiati@unhas.ac.id

[1] Higher Institution Centre of Excellence (HICoE), Institute of Tropical Aquaculture and Fisheries, Universiti Malaysia Terengganu, Kuala Terengganu 21030, Terengganu, Malaysia

[2] Department of Fisheries, Faculty of Marine Science and Fisheries, Hasanuddin University, 90245 Makassar, Indonesia

[3] Department of Fisheries Socio-Economic, Faculty of Fisheries and Marine, Lambung Mangkurat University, 70123 Banjarbaru, Indonesia

[4] Program of Crop Science, Faculty of Food Science and Agrotechnology, Universiti Malaysia Terengganu, Kuala Terengganu 21030, Terengganu, Malaysia

[5] Department of Animal Science and Aquatic Ecology, Faculty of Bioscience Engineering, Ghent University, 9000 Ghent, Belgium

[6] Department of Marine Affairs and Fisheries, East Nusa Tenggara Provincial Government, 85114 Kupang, Indonesia

[7] Faculty of Science and Marine Environment, Universiti Malaysia Terengganu, Kuala Terengganu 21030, Terengganu, Malaysia

[8] Department of Marine Science, Faculty of Marine Science and Fisheries, Hasanuddin University, 90245 Makassar, Indonesia

[9] Centre of Excellence for Seaweed (PUIRL), Institute for Research and Community Service (LPPM), Hasanuddin University, 90245 Makassar, Indonesia

[10] Algal Biotechnology Research Group (AlBioRG), Hasanuddin University, 90245 Makassar, Indonesia